TECHNICAL ANALYSIS & EXPLANATION REPORT

Electricity Management Dashboard

Power Monitor — AOR / GE(N) Maint Isld

> A complete technical analysis of the Power Monitor dashboard: its purpose, architecture, every screen and component, every metric and its formula, the origin and quality of every value it displays, and how the system should be read and interpreted.
>
> This report is written to be read without prior exposure to the dashboard. It documents what the implementation actually does, and states explicitly where a value is measured, derived, estimated or simulated.

SUBJECT SYSTEM	**Power Monitor — Electricity Management Dashboard**
PROJECT	**Power_Consumption_Dashboard**
SOURCE OF RECORD	**AOR - GE (N) Maint Isld.docx**
REPORT PURPOSE	**Technical explanation, handover and review**
TECHNOLOGY	**Vanilla ES modules, zero runtime dependencies, Node.js reference server**
SCOPE ANALYSED	**12 screens · 110 register points · 16 source modules**
REPORT DATE	**4 September 2026**
DOCUMENT STATUS	**Issued for review**

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Provenance markers. Throughout this report, claims carry a marker showing their basis: `IMPL` verified directly from the implementation source; `DOMAIN` standard electrical or SCADA domain knowledge; `INTERP` reasonable interpretation by the analyst. Where something could not be established from the code, the report says so explicitly.

CHAPTER 1

Executive Summary

What the system is, the problem it addresses, and who it serves — written for a reader with no prior exposure to it and no electrical background.

1.1 What the dashboard is

Power Monitor is a web-based electricity management dashboard for the AOR complex at GE(N) Maint Isld — a 360-acre site containing 2,157 buildings, three electrical substations and 109 transformers. It presents, on a single set of screens, how much electrical power every part of the site is drawing, whether any part is drawing more than it should, and what the electrical supply quality looks like.

The dashboard displays 110 monitored locations, each identified by a stable tag such as `TW-01` (Tube Well No.1) or `MED-HOSP` (PNS Hafeez Hospital). Each location has a defined consumption limit. When a location exceeds that limit, the system raises an alarm that persists until an operator acknowledges it. IMPL

1.2 The problem it addresses

Before a system like this exists, a facility of this size has an inventory but no visibility. The maintenance organisation knows it is responsible for 980 street lights, 84 motor pumps and 14 tube wells, but it cannot answer three practical questions:

- **Where is our electricity going?** Which zones, buildings and plant items account for the site's consumption.
- **Is anything abnormal right now?** A pump drawing double its normal current is a failing bearing; nobody notices until it seizes.
- **Do we have capacity?** Whether a new building can be connected to an existing substation without overloading it.

The dashboard is built to answer all three continuously rather than through periodic manual survey.

> CRITICAL QUALIFICATION — READ BEFORE USING ANY FIGURE
>
> The source document for this project is an **asset register**: it records what exists and in what quantity. It contains **no electrical measurements of any kind**. A systematic search of the document returns zero occurrences of *voltage*, *frequency*, *power factor*, *kWh*, *MVAR*, *generator*, *incomer*, *breaker*, *meter* or *demand*. IMPL
>
> Consequently every live reading shown by the dashboard is **simulated**, and is labelled as such in the interface. The structure — asset counts, transformer capacities, named locations, network topology — is real and traceable to the register. The readings are a working model awaiting connection to real metering.

1.3 What information it provides

The dashboard organises information into five categories:

CATEGORY	WHAT IT COVERS	PRIMARY SCREENS
Consumption	Power drawn now, by location, by asset group and by substation; energy accumulated over time	Overview, Load Monitoring, Energy Analytics
Supply quality	Voltage, frequency, power factor, phase imbalance — whether the electricity arriving is within statutory limits	Voltage & Frequency, Overview
Network state	How power flows from intake to load; substation loading; breaker positions	Power Flow, Switchgear, Transformers
Exceptions	Locations over limit, meters not reporting, and a permanent record of every alarm	Alarms & Events, Overview
Planning	Generation from solar, forecast demand, spare capacity, exportable reports	Solar, Forecasting, Reports

Table 1.1 — The five information categories and where each is presented.

1.4 Who would use it

USER	WHAT THEY NEED FROM IT
Control-room operator	Continuous view of whether anything is abnormal now; acknowledge and act on alarms. Uses Overview and Alarms & Events.
Maintenance engineer	Investigate a specific location — its trend, its limits, its alarm history. Uses Load Monitoring and its detail panel.
Electrical engineer	Supply quality, phase imbalance, substation loading, network topology. Uses Voltage & Frequency, Power Flow, Transformers.
Facilities manager	Consumption trends, energy totals, cost drivers, capacity headroom. Uses Energy Analytics, Reports, Forecasting.
Planner / project engineer	Whether spare capacity exists to connect new load. Uses Forecasting and Transformers.

Table 1.2 — Intended audiences and the screens serving each. `INTERP`

1.5 Decisions it can support

- **Immediate response** — dispatch a technician to a location drawing far above its normal profile.
- **Preventive maintenance** — a pump whose consumption is creeping upward week over week is degrading before it fails.
- **Capacity planning** — whether a substation has headroom for additional connected load.
- **Consumption reduction** — the top-consumers table identifies where reduction effort produces the largest result.
- **Fault localisation** — the single-line diagram and power-flow view narrow a problem to a feeder or substation.
- **Data-quality management** — meters that stop reporting are surfaced as a distinct condition from meters reading zero.

A FINDING WORTH ESCALATING

While reconciling the register, one genuine documentary discrepancy was identified. The document states **14 tube wells, of which 12 are operational**, but lists only **11** with locations. One operational tube well is unaccounted for. Tags TW-02, TW-06 and TW-12 have no register entry and are carried as unmonitored. This is a question for the facilities team, and the dashboard displays it rather than silently reconciling it. `IMPL`

CHAPTER 2

Dashboard Overview

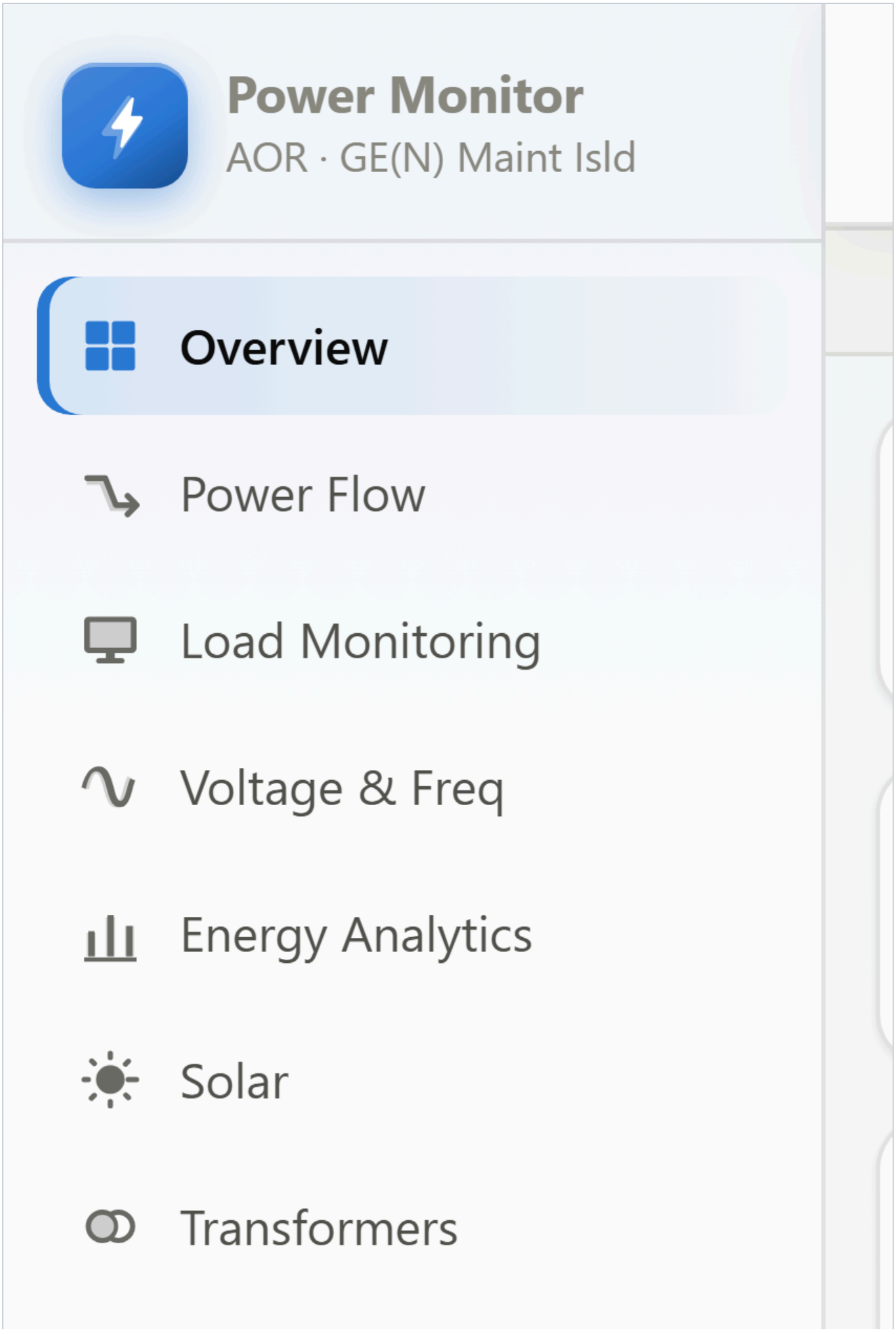
How the dashboard is organised, how its parts relate, and the information hierarchy a reader should expect.

2.1 Organising principle

The dashboard follows a **summary-then-detail** structure. One screen answers "is anything wrong?"; eleven further screens answer "what, where, and why?". This mirrors how a control room is actually used: continuous monitoring of a single overview, with drill-down only when something demands attention. INTERP

2.2 Main navigation

Navigation is a fixed left-hand sidebar listing twelve destinations. It is always visible on screens wider than 900 pixels; below that it becomes a horizontal scrolling strip at the top of the page. IMPL The selected item is marked with a left accent bar and a tinted background. An alarm counter appears against *Alarms & Events* when unacknowledged alarms exist.


Power Monitor
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Overview
Power Flow
Load Monitoring
Voltage & Freq
Energy Analytics
Solar
Transformers

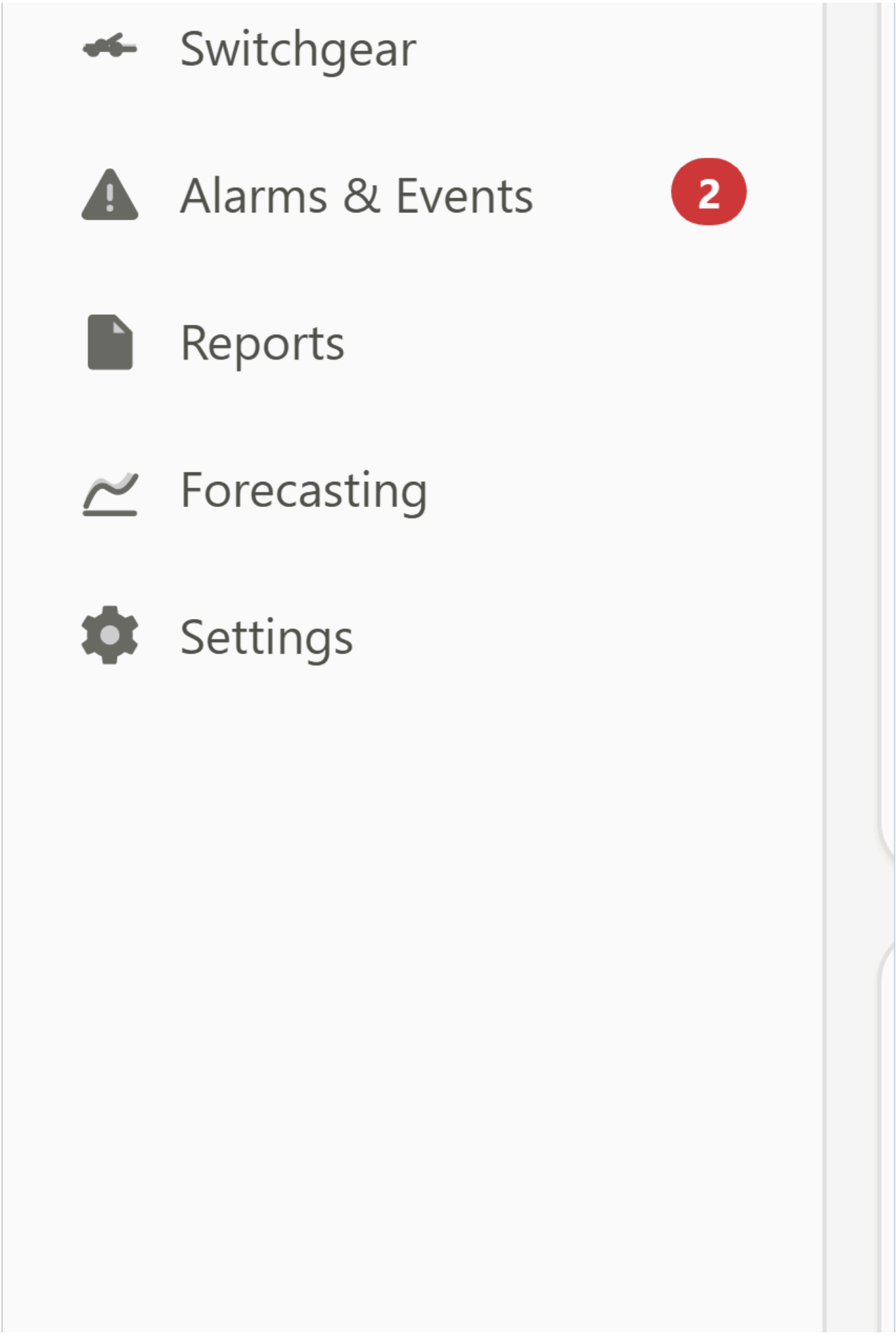

Switchgear
Alarms & Events
2
Reports
Forecasting
Settings

Figure 2.1 — The navigation sidebar. Twelve destinations in a deliberate order: situational awareness first (Overview, Power Flow, Load Monitoring), then supply quality and analysis, then network assets, then exceptions and reporting. The red badge against Alarms & Events is the count of unacknowledged alarms.

SCREEN	CATEGORY	QUESTION IT ANSWERS
Overview	Situational	Is anything wrong right now?
Power Flow	Situational	Where is the electricity going?
Load Monitoring	Situational	What is each individual location drawing?
Voltage & Frequency	Quality	Is the supply itself healthy?
Energy Analytics	Analysis	How much have we consumed, and where?
Solar	Analysis	Is our generation performing as it should?
Transformers	Assets	What transformer capacity do we have?
Switchgear	Assets	What is the state of the protective equipment?
Alarms & Events	Exceptions	What has gone wrong, and was it handled?
Reports	Reporting	What are the period totals, and can I export them?
Forecasting	Planning	What will demand be over the next day?
Settings	Reference	Where does every number come from?

Table 2.1 — The twelve screens, in navigation order. IMPL

2.3 Persistent header



Figure 2.2 — The persistent header. From left: title and site identifier, live indicator, clock and date, connection health pill, data-source selector, pause control, alarm bell with count, display-size toggle, and theme toggle.

The header remains constant across all twelve screens. Its most operationally significant element is the **connection pill**, which reports both the feed state and the age of the most recent message. A dashboard that silently freezes on stale values is the most dangerous failure mode this class of system has; the header is where that condition is made visible. INTERP

2.4 Provenance bar

Figure 2.3 — The provenance bar, displayed beneath the header on every screen. It defines the two badges that appear throughout the interface: **REG** for values taken from the asset register, **SIM** for simulated readings.

This bar is unusual for a dashboard and deserves comment. Because the register contains no metering, the interface would otherwise present modelled values with the same authority as measured ones. The two-badge scheme makes the distinction visible at the point of use rather than burying it in documentation. IMPL

2.5 Information hierarchy

Within the Overview screen, and repeated on the detail screens, the hierarchy runs:

```
Level 1   Headline figures        6 KPI tiles / 4 stat tiles
          "what is the state?"    single numbers, largest type on the page

Level 2   Trends and breakdowns   charts, distribution, category bars
          "how did we get here?"  shape over time, share by group

Level 3   Exceptions              alarm list, status badges
          "what needs action?"    only populated when something is wrong

Level 4   Detail                  tables, per-point drill-down
          "what exactly?"         every location, every value, every timestamp
```

Figure 2.4 — The four-level information hierarchy applied across the dashboard. INTERP

2.6 How the parts work together

All twelve screens read from the same three in-memory models, so no two screens can disagree about the state of the plant: IMPL

- **The store** holds smoothed readings, per-point history, and roll-ups by group and by substation.
- **The electrical model** derives voltage, current, power factor, reactive power and energy from those readings.

- **The alarm engine** evaluates every reading against its limits and maintains the alarm record.

A single render loop updates whichever screen is currently displayed, once per second. Screens that are not displayed are not rendered.

CHAPTER 3

System & Technical Architecture

The implementation as it actually exists: languages, modules, runtime behaviour and dependencies.

3.1 Technology summary

LAYER	TECHNOLOGY	NOTES
Frontend framework	None — vanilla JavaScript	Native ES modules loaded by `<script type="module">`. No React, Vue, Angular or Svelte. IMPL
Visualisation library	None — hand-authored SVG	All charts are generated as SVG by project code. No D3, Chart.js, Recharts or Plotly. IMPL
Styling	Plain CSS in `index.html`	CSS custom properties drive two themes. No Tailwind, Bootstrap or preprocessor.
Backend	Node.js `node:http`	Reference server in `server/mock-scada.js`. No Express, Fastify or Koa. IMPL
Database	None	No database of any kind. State is in browser memory and is not persisted across reloads, except UI preferences. IMPL
Runtime dependencies	Zero	`package.json` declares no `dependencies` or `devDependencies`. Nothing to `npm install`. IMPL
State management	Module closures	Each model is a factory function returning an object with private state. No Redux, MobX or signals.
Authentication	None	No login, session or access control exists in the implementation. IMPL
External services	None	No CDN, analytics, telemetry or third-party API calls. The application runs fully offline. IMPL
Node requirement	≥ 18	Declared in `package.json` `engines`.

Table 3.1 — Technology stack, verified against the project files.

> **WHY ZERO DEPENDENCIES MATTERS HERE**
>
> A facility of this type may operate on an isolated or air-gapped network. A build with no external dependencies, no CDN references and no build step can be deployed by copying a folder, and cannot break because a package registry is unreachable. INTERP

3.2 Module structure

MODULE	LINES	RESPONSIBILITY
`src/register.js`	404	The point register — every monitored location, its tag, group, substation, unit count, rated demand and limits. The single source of truth for what is monitored.

MODULE	LINES	RESPONSIBILITY
src/source.js	238	Data-source abstraction. Three modes: in-browser simulator, server-sent events, REST polling. Includes the connection watchdog. **This is the sole integration seam.**
src/simulator.js	190	Stand-in telemetry generator: diurnal demand profiles, mean-reverting noise, load excursions, comms dropouts.
src/store.js	271	Reading history, exponential smoothing, energy accumulation, and roll-ups by group and substation.
src/electrical.js	200	Derives voltage, current, power factor, reactive and apparent power, energy, losses and CO_2 from real power.
src/limits.js	434	The alarm engine: limit evaluation, on/off delays, hysteresis, latching, acknowledgement, anti-flood re-arm, episode log.
src/switchgear.js	131	Breaker, feeder and distribution-board state model.
src/overview.js	~290	Renders the Overview screen and the Transformers screen.
src/pages.js	~510	Renders Power Flow, Voltage & Frequency, Energy Analytics, Solar, Switchgear and Forecasting.
src/ui.js	843	Point table, group cards, alarm lists, detail drawer, Reports screen, CSV export, connection banner.
src/charts.js	372	Trend chart, sparkline, horizontal bar chart.
src/charts-power.js	~330	Distribution doughnut, arc gauge, multi-series chart, single-line diagram.
src/charts-pages.js	~330	Power-flow diagram, status grid, forecast band chart, load-duration curve.
src/format.js	84	Number, unit, percentage and time formatting.
src/app.js	222	Wiring: instantiates the models, binds navigation, runs the render loop.
server/mock-scada.js	~245	Reference server: serves the static application and four API endpoints.

Table 3.2 — Source modules and responsibilities. IMPL

3.3 API endpoints

ENDPOINT	METHOD	RETURNS
/api/readings	GET	The most recent sample set for every reporting point. Used by polling clients.
/api/stream	GET (SSE)	Server-sent event stream, one frame per tick. Used by streaming clients. Primes immediately on connect and sends a keep-alive comment every 15 s.
/api/points	GET	The full point register: tags, names, locations, groups, substations, unit counts, ratings and limits.
/api/health	GET	Tick count, uptime, configured interval, connected stream clients, monitored point count, points reporting in the last tick.

Table 3.3 — API surface of the reference server. IMPL

3.4 Runtime behaviour

Feed interval	2,000 ms default; configurable via `--interval`.
Render interval	1,000 ms, fixed and independent of the feed rate.
Source detection	On load the application probes `./api/health` with a 2.5 s timeout. If it responds, the stream is used; otherwise the in-browser simulator runs.
Persistence	`localStorage` retains only theme, display density and last-viewed screen. No readings are persisted.

> **DESIGN NOTE — WHY RENDERING IS DECOUPLED FROM THE FEED**
>
> The render loop runs on a fixed timer rather than on data arrival. A fast feed therefore cannot overwhelm the interface, and — more importantly — a feed that *stops* still triggers repaints. This is what allows staleness, offline meters and a dead backend to surface on their own, instead of the display freezing on the last good value while appearing healthy. `IMPL`

3.5 Verification tooling

Two scripts accompany the implementation: `IMPL`

- `scripts/verify-register.js` — reconciles the point register against the totals stated in the source document (23 assertions covering building counts, asset counts, transformer units and installed capacity, plus limit-ladder coherence and duplicate-tag detection).
- `scripts/test-alarms.js` — 26 behavioural assertions on the alarm engine covering on-delay, anti-flood, hysteresis, acknowledgement, ringback, re-arm, comms detection and smoothing.

Both run via `npm test`. All 29 checks pass on the analysed revision.

CHAPTER 4

Data Flow

The path a single reading takes from origin to pixel, and where each transformation occurs.

4.1 End-to-end pipeline

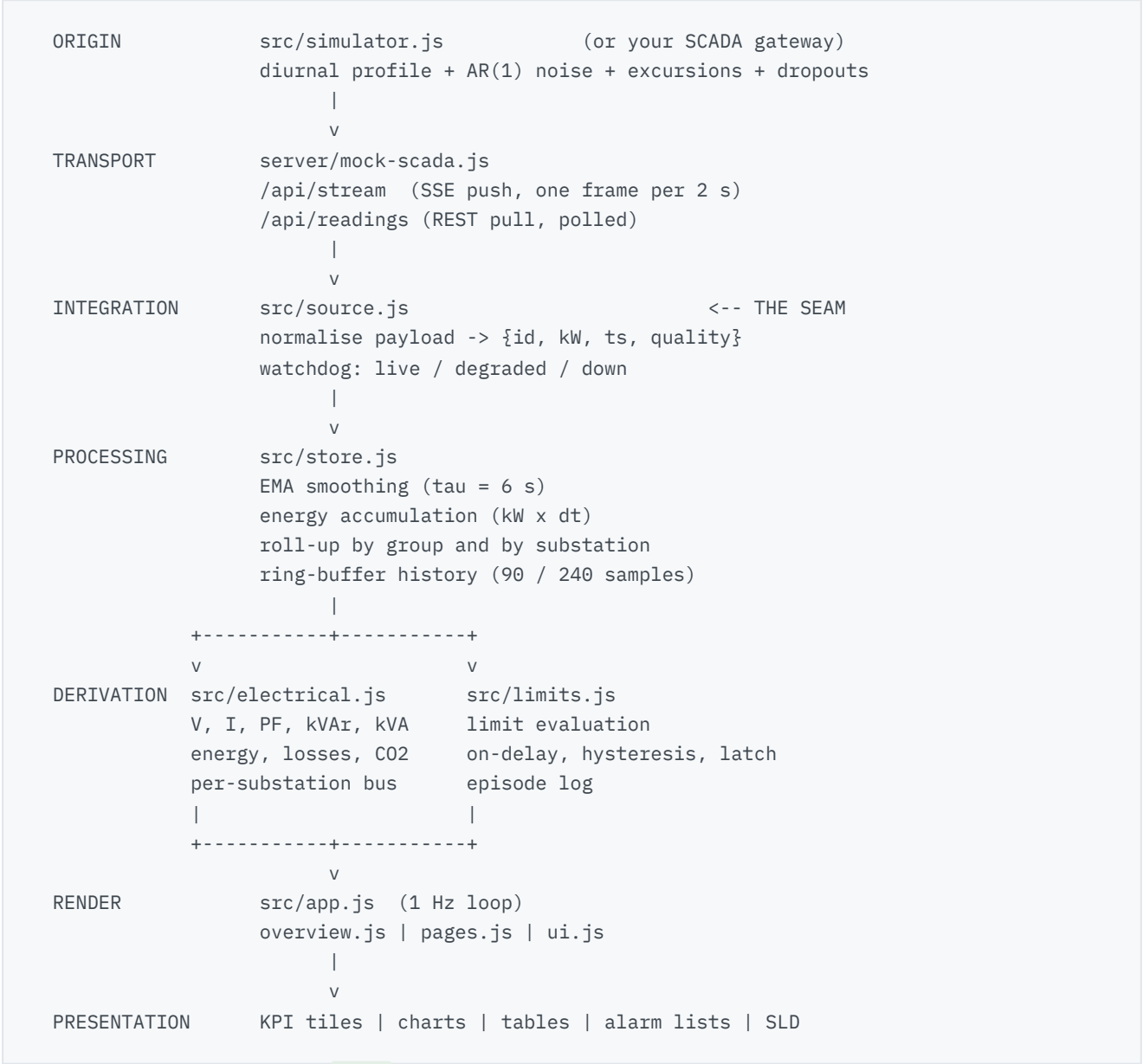


Figure 4.1 — The complete data pipeline. IMPL

4.2 Stage detail

STAGE	MODULE	TRANSFORMATION APPLIED
Origin	`simulator.js`	Generates a kW value per point from its group's diurnal profile, plus mean-reverting noise, occasional excursions and comms dropouts. Points in dropout are *omitted* from the payload, never sent as zero.

STAGE	MODULE	TRANSFORMATION APPLIED
Transport	`mock-scada.js`	Serialises the sample set to JSON and pushes it over SSE or serves it on request.
Integration	`source.js`	Normalises field-name variants, coerces numerics, discards malformed rows, and tracks feed health independently of content.
Smoothing	`store.js`	Exponential moving average with a 6-second time constant. The raw sample is retained alongside.
Accumulation	`store.js`	Integrates smoothed power over elapsed time into energy, per point and site-wide.
Aggregation	`store.js`	Sums points into group and substation roll-ups on every ingest; generation is signed negative in load groups but positive in the generation group.
Derivation	`electrical.js`	Computes all electrical quantities from real power using standard three-phase relationships.
Evaluation	`limits.js`	Compares the smoothed value against warning and alarm limits with timing and hysteresis rules.
Render	`app.js` + views	Writes only the DOM cells whose value changed. The displayed screen is rendered; others are not.

Table 4.1 — Transformation applied at each pipeline stage. IMPL

4.3 The payload contract

Both transports carry an identical structure. This is the contract a real SCADA gateway must satisfy: IMPL

```
{
  "ts": 1785755194099,          // epoch ms, when the frame was assembled
  "daylight": true,             // optional; gates solar under-performance limits
  "readings": [
    { "id": "TW-01",            // must match a tag in register.js
      "kW": 14.8,               // instantaneous demand; generation is POSITIVE
      "ts": 1785755194099,      // when sampled AT THE METER, not on arrival
      "quality": "good" }       // "good" | "bad"
  ]
}
```

> **CONTRACT RULE — UNREACHABLE METERS**
>
> A meter that cannot be reached must be **omitted from the readings array entirely**. It must never be sent as `0`. The dashboard distinguishes "offline" from "drawing nothing", and transmitting zeros would destroy that distinction — a genuinely dead circuit and a failed communications link would become indistinguishable. IMPL

4.4 Connecting real telemetry

Because every screen reads from the models rather than the feed, replacing simulated data with real data requires changing one module. In `src/app.js` the source is constructed with a mode and two URLs; pointing those at a real gateway is sufficient. The source mode is also switchable at runtime from the header. IMPL

The reader is tolerant of common field-name variants — `tag` or `pointId` for `id`, `kw` or `value` for `kW`, `timestamp` for `ts` — which reduces the amount of adaptation a gateway needs. IMPL

CHAPTER 5

Screen Analysis — Overview

The first of twelve screen analyses. Each follows the same structure: purpose, rationale, information provided, mechanism, interpretation, and a practical use case. Every major section within the screen is documented individually.

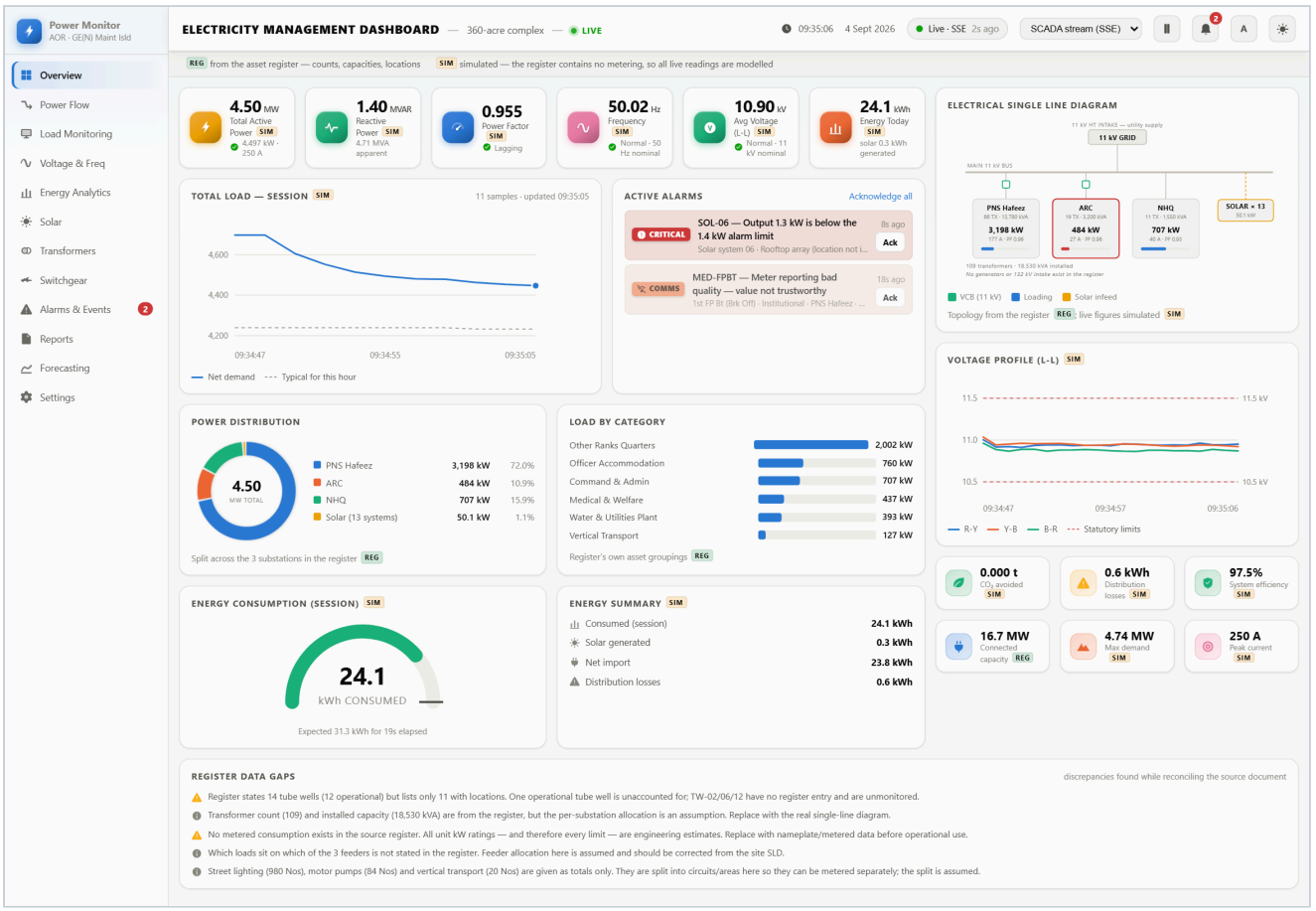


Figure 5.1 — The Overview screen in full. Left column: KPI strip, load trend, active alarms, and the four analysis panels. Right rail: single-line diagram, voltage profile, secondary metrics. Full width at the foot: register data gaps.

5.1 Purpose

To answer one question without scrolling and without interaction: **is the site healthy right now?** This is the screen intended to remain permanently displayed on a wall-mounted monitor.

5.2 Why this screen exists

Continuous monitoring only works if the normal state is legible at a glance from across a room. If an operator must read numbers or navigate to establish that nothing is wrong, the monitoring is not continuous — it is periodic inspection. The Overview compresses six headline electrical parameters, the alarm state, the network diagram and the supply-quality trace into a single view. `INTERP`

5.3 Sections

5.3.1 KPI strip

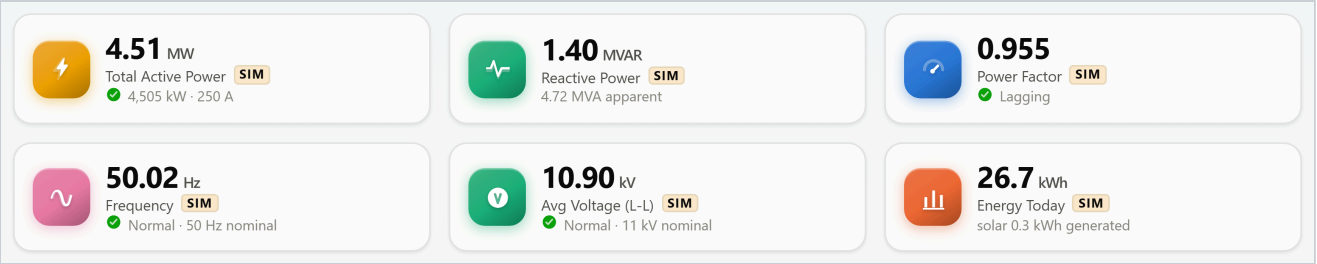


Figure 5.2 — The six headline KPI tiles. Each carries an icon medallion, the value with its unit, a label with its provenance badge, and a status line with a tick or warning glyph indicating whether the parameter is within its normal band.

Purpose	Present the six parameters that together characterise the electrical state of the site.
Contents	Total Active Power (MW) · Reactive Power (MVAR) · Power Factor · Frequency (Hz) · Average Voltage L-L (kV) · Energy Today (kWh/MWh).
Data used	All six read from the electrical model, which derives them from the summed real power of all monitored points. `IMPL`
Relationship	These are site-wide aggregates of the same readings broken down on Load Monitoring and analysed on Energy Analytics.
Interpretation	Four tiles carry a pass/fail indicator: power factor (green at ≥ 0.90), frequency (green within 49.5–50.5 Hz), voltage (green within 10.5–11.5 kV). If all show green, supply quality is acceptable.
Significance	These are the figures an electrical engineer asks for first. Full definitions appear in Chapter 17.

5.3.2 Total load trend

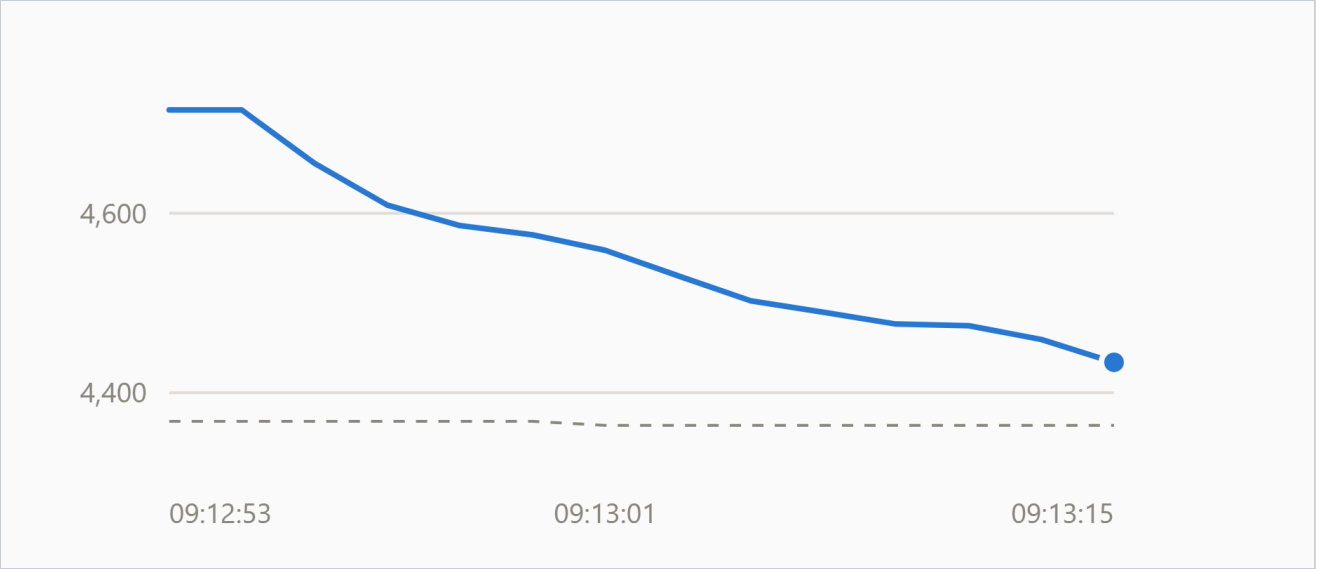


Figure 5.3 — Site demand over the session, with the dashed grey line showing typical demand for the current hour. The y-axis is zoomed to the data range rather than anchored at zero, so variation of a few hundred kW on a 4,500 kW base remains visible.

Purpose	Show the shape of demand over time and whether the site is above or below its normal profile.
Data used	The site history ring buffer, holding up to 240 samples of net demand and expected demand. `IMPL`
Unit	kW on the y-axis; clock time on the x-axis.

Interpretation	The solid line tracking near the dashed line means the site is behaving normally for the time of day. A sustained gap above it is the signal worth investigating — a raw value alone cannot tell you that.
Significance	Provides the context that turns a number into a judgement.

5.3.3 Active alarms

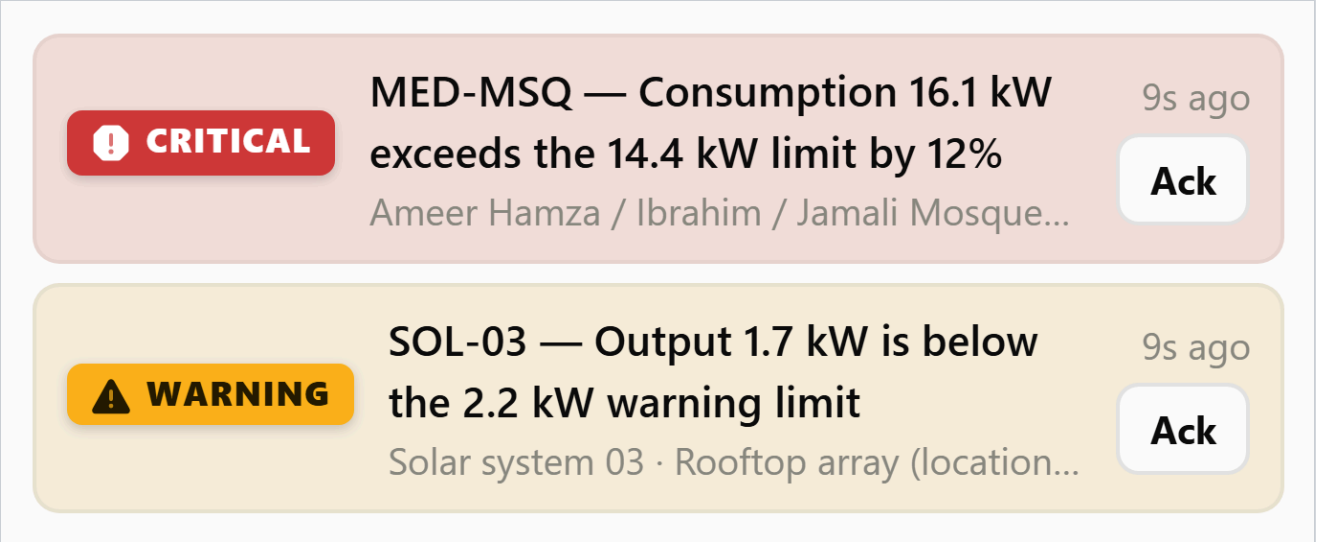


Figure 5.4 — The active alarm panel. Each row carries a severity tag, the affected tag and message, the location and feeding substation, the elapsed time since it was raised, and an acknowledge button.

Purpose	Surface every condition currently requiring operator attention, worst first.
Data used	The alarm engine's open-episode list, sorted by severity, then by acknowledgement state, then by recency. `IMPL`
Interpretation	A **Critical** tag means consumption exceeded the alarm limit. **Warning** means the warning limit. **Comms** means the meter stopped reporting — a communications fault, not an electrical one. When empty it states so explicitly rather than showing nothing.
Significance	This panel is the operational core of the screen; everything else is context for it.

5.3.4 Power distribution

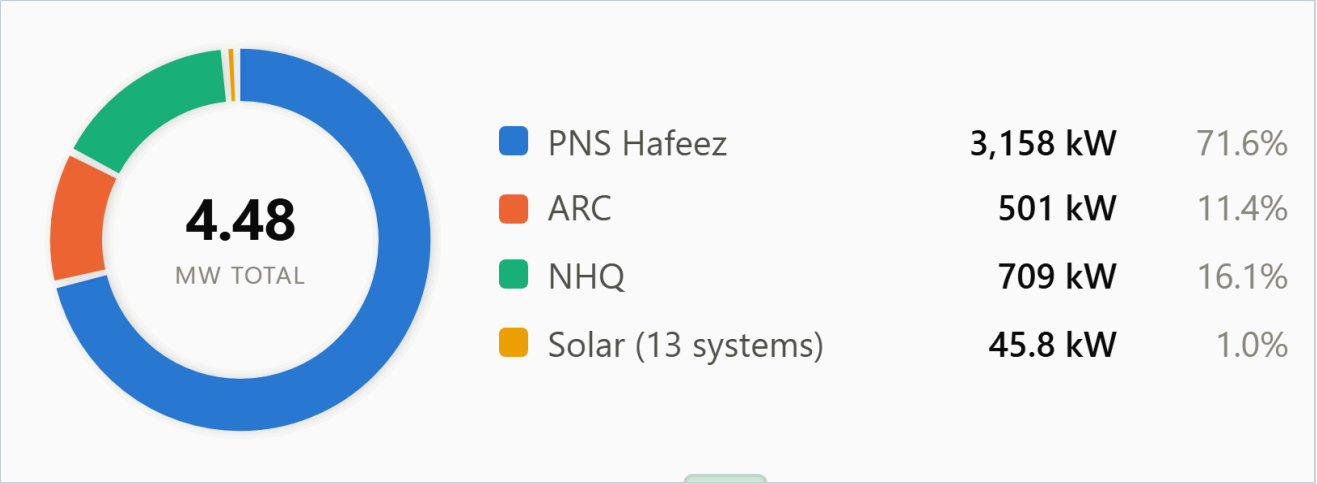


Figure 5.5 — Demand split across the three substations plus solar infeed, with the site total at the centre. Each segment is direct-labelled with its value and share, so identity never depends on colour alone.

Purpose	Show how site demand divides between supply points.

Data used	Substation roll-ups from the store; solar from the generation roll-up. IMPL
Unit	kW and percentage of total.
Relationship	Same underlying figures as the substation boxes on the single-line diagram and the feeder bars on Power Flow.
Interpretation	A structural view rather than a diagnostic one. It is expected to be stable; a sudden change in the split means load has transferred or a substation has lost supply.

5.3.5 Load by category

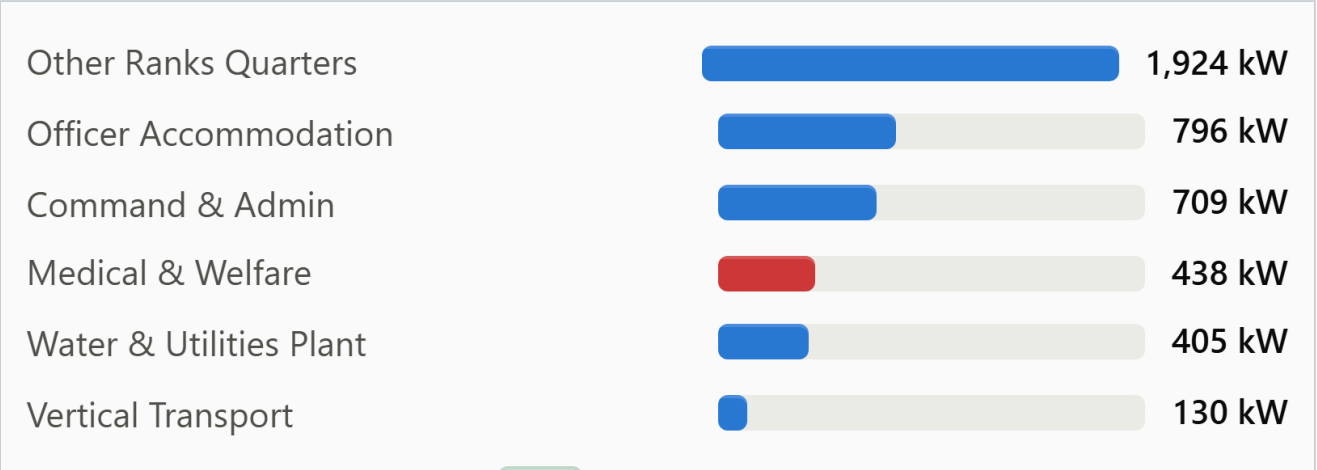


Figure 5.6 — The six largest asset groups by current demand. A bar turns amber or red when that group holds a warning or critical alarm — colour here carries alarm state, not magnitude.

Purpose	Show which categories of asset are consuming the most.
Data used	Group roll-ups, sorted descending, top six shown. IMPL
Grouping	Uses the register's own taxonomy — Officer Accommodation, Other Ranks Quarters, Command & Admin, Medical & Welfare, Water & Utilities, Street & Area Lighting, Vertical Transport, Outstations, Solar Generation — rather than inventing new categories.
Interpretation	Residential groups dominate, which is expected for a complex of 2,157 buildings. Rank order changing through the day is normal; a group jumping rank abruptly is not.

5.3.6 Energy consumption gauge

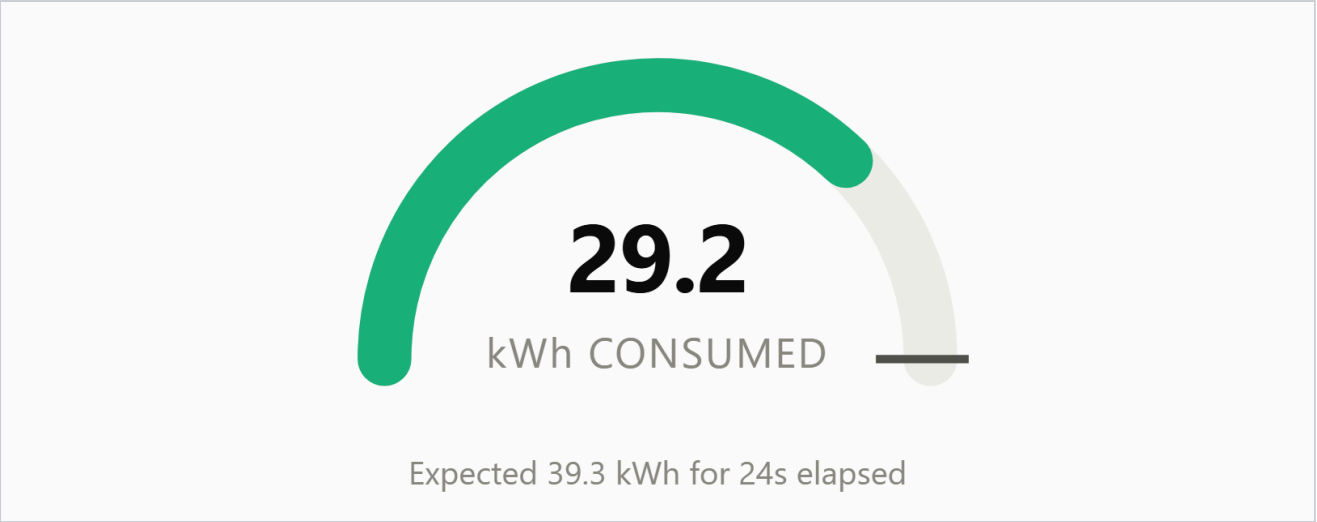


Figure 5.7 — Energy consumed this session against the amount expected for the elapsed period. The tick mark on the arc is the expected value; the caption states it explicitly.

Purpose	Compare accumulated energy against a reference for the same elapsed time.
Calculation	Reference = site rated demand × elapsed hours × 0.85. The register contains no consumption target, so this reference is derived rather than given. `IMPL`
Unit	kWh below 1,000; MWh above.
Interpretation	Fill beyond the tick means consumption is running ahead of expectation for the period. The arc turns amber above 85% and red above 95% of the reference.

5.3.7 Energy summary



Figure 5.8 — Four accumulated energy figures: consumed, solar generated, net import, and distribution losses.

Purpose	Give the energy balance in four lines.
Calculation	Net import = consumed − solar generated. Losses accumulate at a demand-dependent fraction (Chapter 19). `IMPL`
Relationship	These four figures reappear as the doughnut segments on Energy Analytics.

5.3.8 Electrical single-line diagram

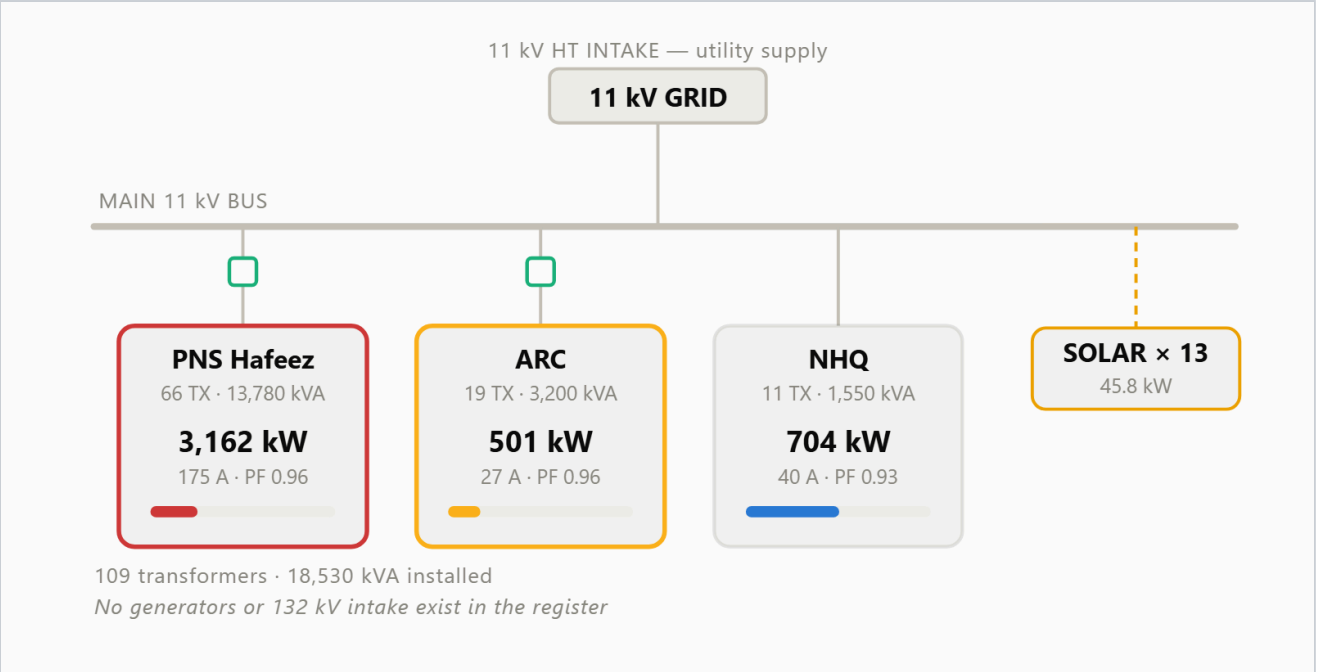


Figure 5.9 — The 11 kV network: grid intake, main bus, the three substations with live load, current, power factor and loading bar, and the solar infeed on a reserved right-hand branch. The note beneath states that no generators or 132 kV intake exist in the register.

Purpose	Present the network in the form an electrical engineer already thinks in, so no mental translation from a generic chart is required.
Data used	Topology and transformer counts from the register; live load, current and power factor from the electrical model. `IMPL`
Status encoding	A substation box outlines amber or red when any point it feeds holds an alarm — linking the diagram directly to the alarm list.
Interpretation	Green squares are the VCBs recorded in the register. The loading bar under each substation shows demand against installed transformer capacity.
Significance	The only screen element that shows how the plant is actually connected.

5.3.9 Voltage profile

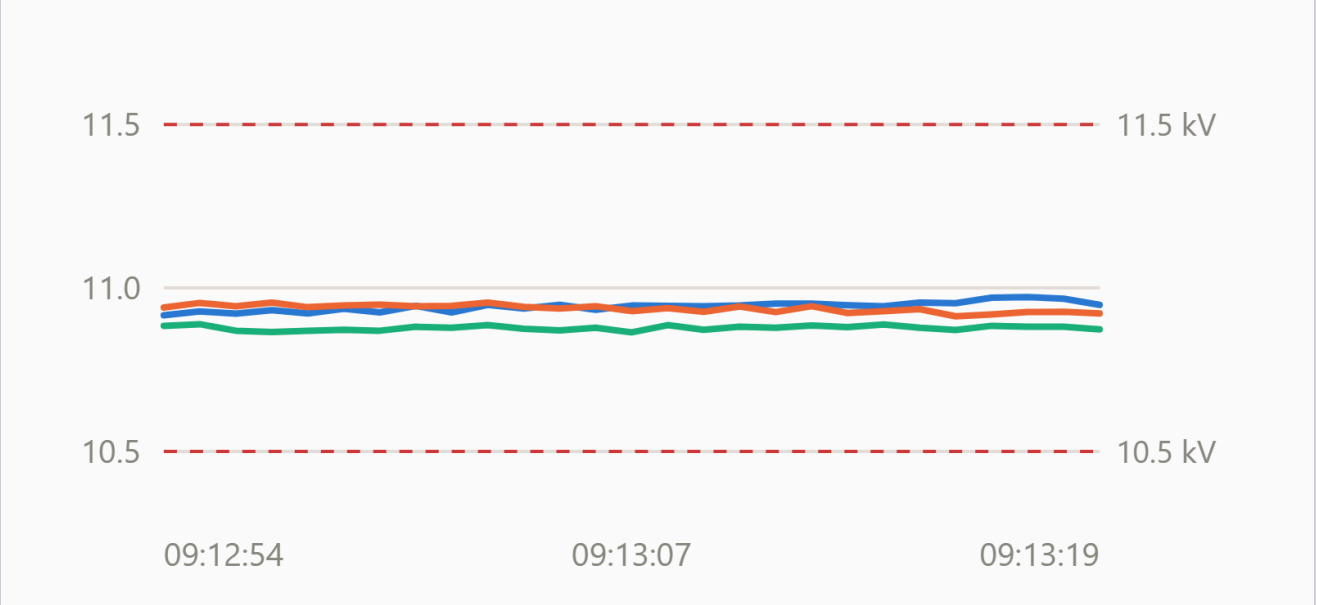


Figure 5.10 — All three line-to-line phase voltages over time, with the 10.5 kV and 11.5 kV statutory limits drawn as red dashed reference lines.

Purpose	Show supply quality alongside supply quantity.
Unit	kV, two decimal places.
Interpretation	Three lines tracking closely together is healthy. Lines separating indicates phase imbalance; the full analysis is on the Voltage & Frequency screen.

5.3.10 Secondary metric tiles

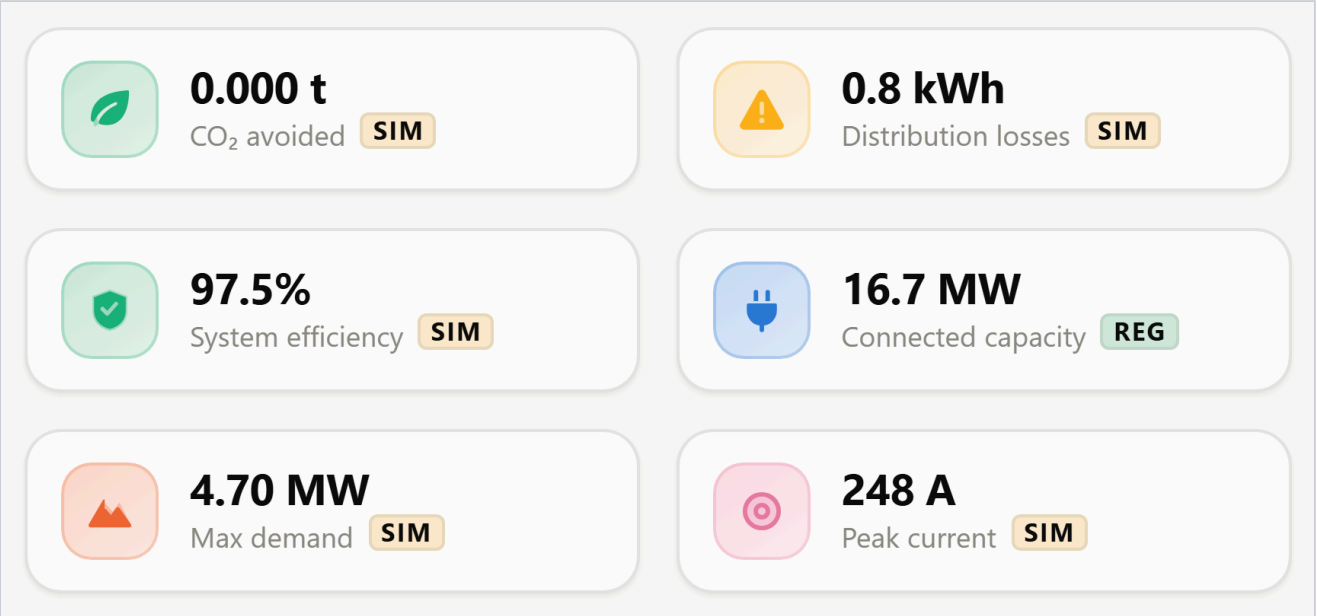


Figure 5.11 — Six secondary metrics. Note the provenance badges: connected capacity is **REG** because it derives from register transformer data; the other five are **SIM**.

Contents	CO_2 avoided (t) · distribution losses (kWh) · system efficiency (%) · connected capacity (MW) · max demand (MW) · peak current (A).
Purpose	Carry the figures that matter periodically but do not warrant permanent headline status.

5.3.11 Register data gaps

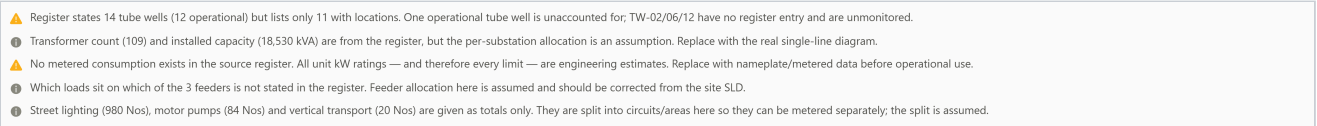


Figure 5.12 — Five documented discrepancies found while reconciling the register, displayed on the operational screen rather than confined to documentation.

Purpose	Make the limits of the underlying data visible at the point of use.
Contents	The tube-well count discrepancy; the assumed per-substation transformer allocation; the absence of any metered consumption; the unstated feeder allocation; and the assumed circuit splits for lighting, pumps and vertical transport. `IMPL`
Significance	An unusual inclusion. Its effect is that a reader cannot mistake an assumption for a measurement without having been told.

5.4 How it works

All Overview content is rendered by `src/overview.js`, called once per second from the render loop while this screen is displayed. Values are read from the store, the electrical model and the alarm engine; the screen performs no data acquisition of its own. DOM writes are made only where a value has changed, so scroll position and keyboard focus survive updates. `IMPL`

5.5 How to interpret it

1. Check the four status indicators in the KPI strip. Green across means supply quality is acceptable.
2. Check the alarm panel. Empty means no location is over limit.
3. Glance at the load trend — is the solid line near the dashed one?
4. Glance at the single-line diagram — are all three substation boxes outlined neutrally?

If all four checks pass, no further inspection is warranted.

5.6 Practical use case

An operator on shift notices the alarm panel now shows one amber entry: *ADM-TWR-C6C7 — consumption 267 kW exceeds the 246 kW warning limit by 9%*. Glancing at the single-line diagram, the NHQ substation box is outlined amber, confirming the affected load sits on that substation. The KPI strip still shows frequency and voltage normal, so this is a load problem rather than a supply problem. The operator acknowledges the alarm and moves to Load Monitoring to examine the tower's trend.

CHAPTER 6

Screen Analysis — Power Flow

Where the electricity goes, traced from intake to consuming group.

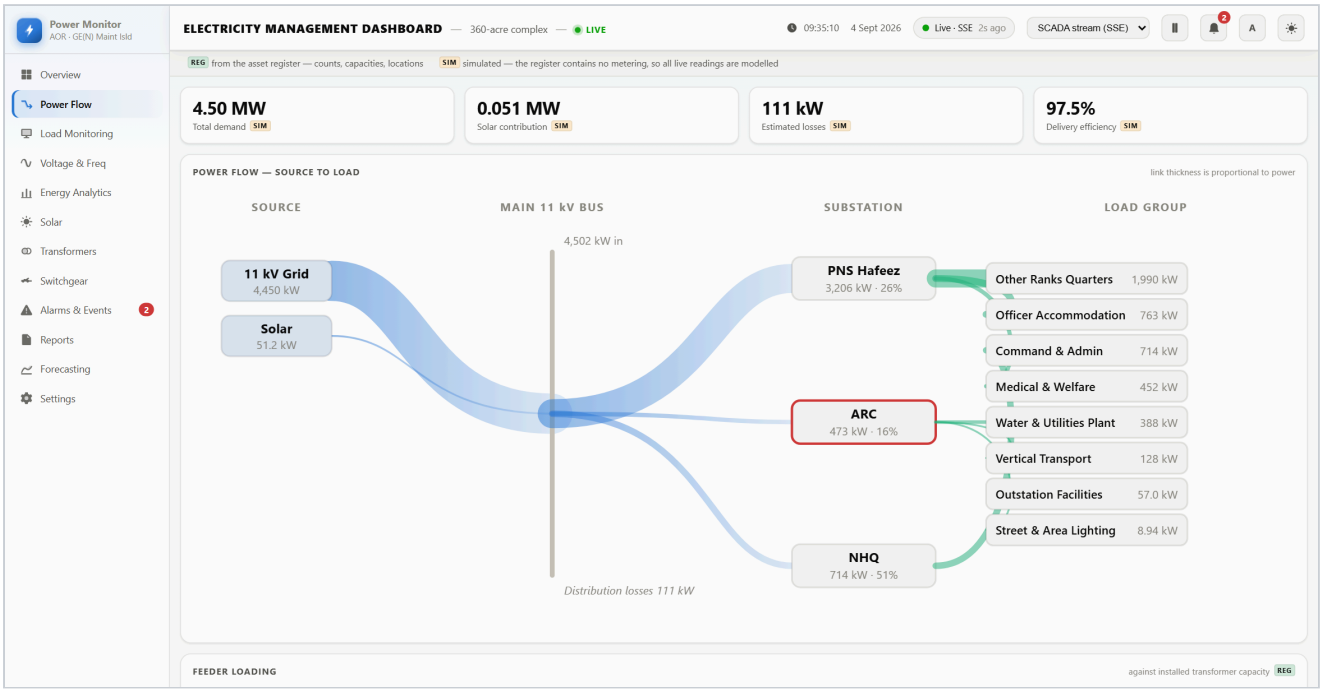


Figure 6.1 — The Power Flow screen: four stat tiles, the flow diagram, and feeder loading.

6.1 Purpose

To show the distribution of power from its sources, through the 11 kV bus and the three substations, to the asset groups that consume it — with quantities represented proportionally rather than only stated numerically.

6.2 Why this screen exists

A table of group consumption tells you the numbers but not the structure. A flow diagram shows both simultaneously: which substation feeds which groups, and how much of the total each path carries. For explaining the site to someone unfamiliar with it, and for narrowing a fault to a supply path, the structural view is faster than any table. `INTERP`

6.3 Sections

6.3.1 Stat tiles

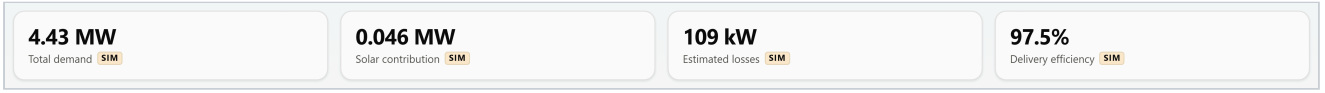


Figure 6.2 — Total demand, solar contribution, estimated losses and delivery efficiency.

Purpose	Quantify the flow in four figures before the diagram is read.

Calculation	Losses = total demand × (100 − efficiency) ÷ 100. Efficiency derives from the demand-dependent loss fraction in the electrical model. `IMPL`
Unit	MW for demand and solar, kW for losses, percent for efficiency.

6.3.2 Flow diagram

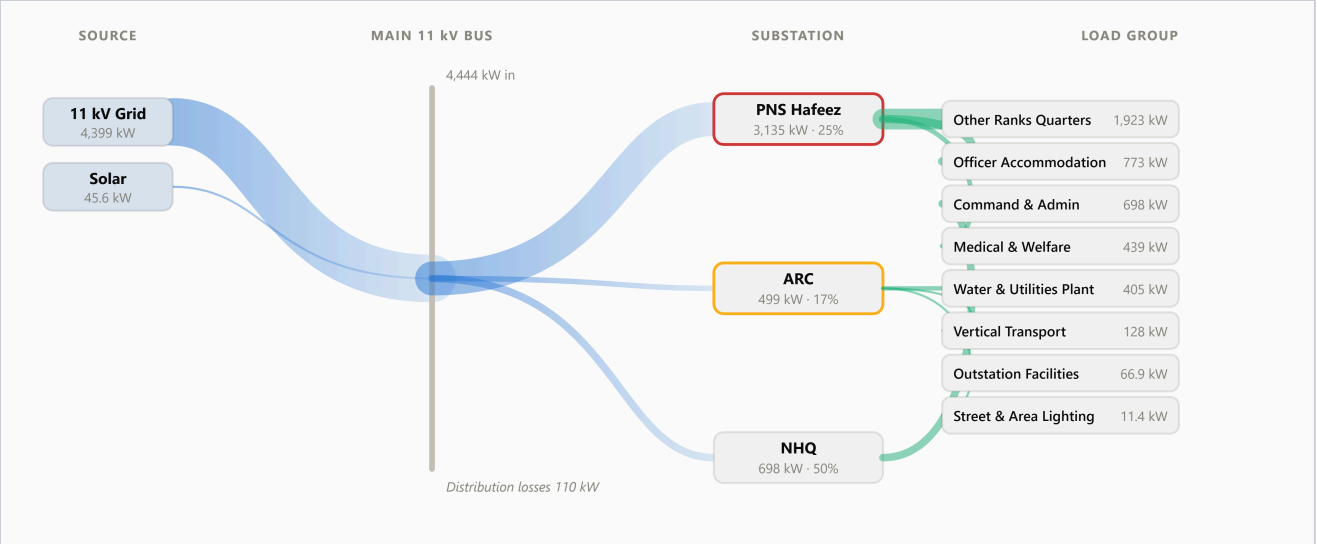


Figure 6.3 — The flow diagram. Four columns left to right: source, main 11 kV bus, substation, load group. Ribbon thickness is proportional to the power carried; ribbons entering the load groups are drawn in a second hue so the final hop is distinguishable.

Purpose	Represent power distribution as proportional flow.
Data used	Sources from site load and generation; substation and group values from store roll-ups; each substation links only to groups that actually contain points fed by it. `IMPL`
Visualisation logic	Ribbon thickness = (value ÷ total input) × 34 px, with a 1.5 px floor so small flows remain visible. `IMPL`
Interpretation	Follow any ribbon from left to right to see the complete path. Thickness comparison between ribbons is the intended reading; exact values are labelled on each node.
Significance	Makes the dominance of residential load immediately apparent without arithmetic.

6.3.3 Feeder loading



Figure 6.4 — Each of the three feeders as a percentage of installed transformer capacity, with the absolute load and capacity stated.

Purpose	Show how hard each feeder is working relative to what it can carry.
Calculation	Loading % = feeder load ÷ substation capacity × 100, where capacity = installed kVA × 0.9. `IMPL`
Status	Amber above 80%, red above 90%.
Interpretation	Typical values sit between 20% and 45%, which is normal distribution-transformer loading. Sustained values above 80% indicate a capacity constraint.

6.4 How to interpret it

Read left to right. The two source nodes should sum to the bus figure. Substation ribbons should be proportionate to the substation loads shown elsewhere. Any substation whose box is outlined amber or red holds an alarm on one of its loads.

6.5 Practical use case

A planner is asked whether a new accommodation block can be fed from the ARC substation. The feeder bars show ARC running at roughly 20% of its 2,880 kW capacity — approximately 2,300 kW of headroom. The flow diagram confirms ARC currently feeds the water, lighting and vertical-transport groups, so the new block would introduce a residential profile onto a feeder that presently carries none, changing its daily shape.

CHAPTER 7

Screen Analysis — Load Monitoring

The complete working list of every monitored location, and the drill-down an engineer uses to investigate one of them.

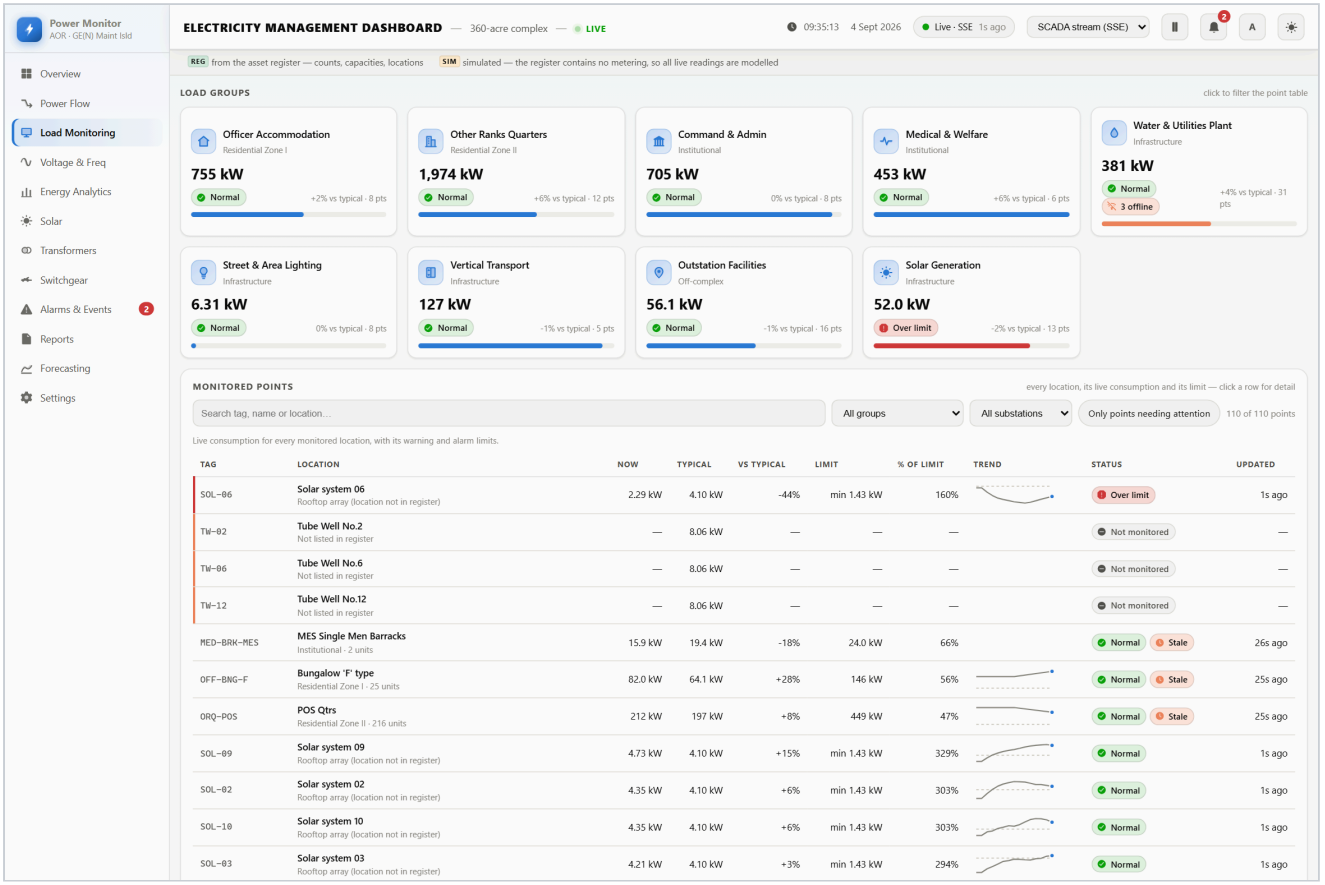


Figure 7.1 — The Load Monitoring screen: nine load-group cards above the filter row and the point table.

7.1 Purpose

To expose every individual monitored location with its live consumption, its limits, its trend and its status — searchable, filterable and sortable.

7.2 Why this screen exists

The Overview establishes that something is wrong; this screen establishes exactly what and where. It is the working surface for anyone investigating a specific asset, and the only screen presenting all 110 register entries at once. `INTERP`

7.3 Sections

7.3.1 Load group cards

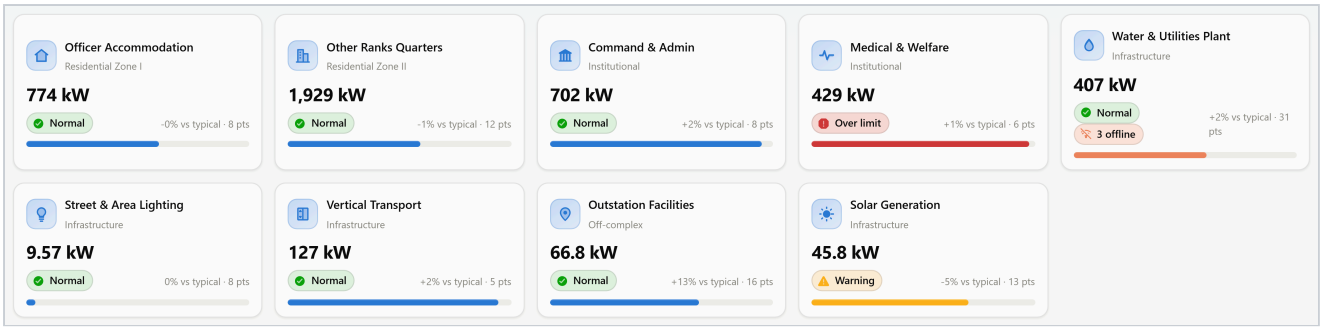


Figure 7.2 — Nine group cards. Each shows the group name and zone, current demand, a status badge with any offline count, deviation from typical, point count, and a loading bar.

Purpose	Give a per-group summary and act as a one-click filter for the table below.
Data used	Group roll-ups; worst severity among the group's points; count of points offline. `IMPL`
Behaviour	Clicking a card filters the table to that group and scrolls to it. Clicking again clears the filter without scrolling. `IMPL`
Interpretation	Severity and communications state are shown side by side, never one instead of the other — a group holding a critical alarm must not read as merely "offline" because one of its meters is also unreachable.

7.3.2 Filter row

Figure 7.3 — The filter row: free-text search, group selector, substation selector, an attention-only toggle, and a live result count.

Filters are covered in full in Chapter 15. In summary: all four combine with AND logic, and the result count updates live. `IMPL`

7.3.3 The point table

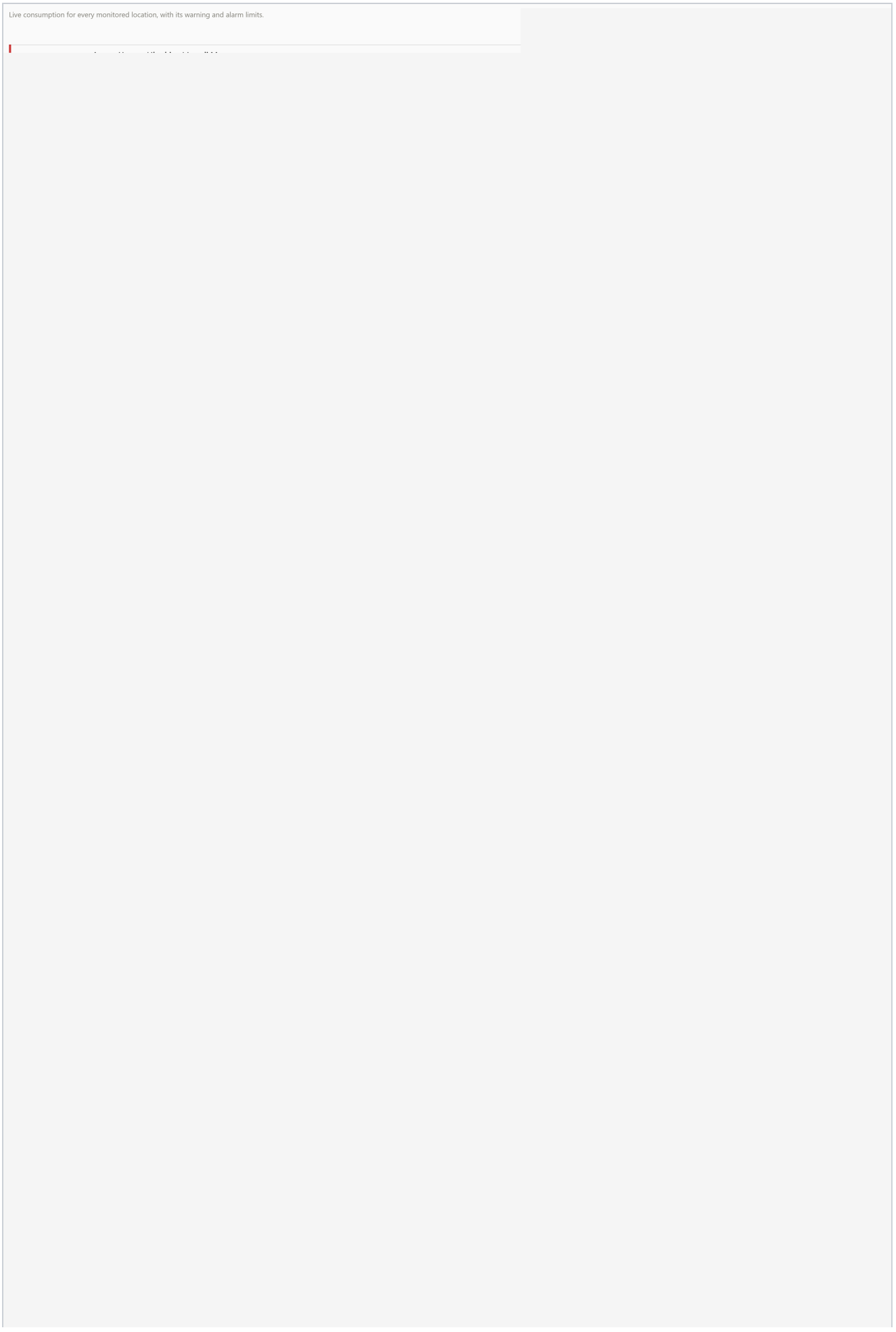

Live consumption for every monitored location, with its warning and alarm limits.

Figure 7.4 — The point table. Rows carry a coloured left edge for warning, critical or communications state. Unmonitored register entries (TW-02, TW-06, TW-12) show em-dashes with a "Not monitored" badge rather than zeros.

COLUMN	UNIT	MEANING AND HOW IT IS DERIVED
Tag	—	The point identifier, keyed to the register. Monospaced for scanning.
Location	—	Asset name, with physical location and unit count beneath.
Now	kW	The smoothed current reading. Shows an em-dash if the meter is offline or the point is unmonitored.
Typical	kW	Expected demand for this hour: rated demand × the group's diurnal profile factor.
vs typical	%	Signed deviation: (now ÷ typical) − 1.
Limit	kW	The alarm limit. Prefixed "min" for generation points, where the limit is a floor rather than a ceiling.
% of limit	%	now ÷ limit × 100. The single most scannable column for proximity to alarm.
Trend	—	Sparkline of the last 24 samples with a reference line at typical.
Status	—	Icon, word and colour: Normal, Warning, Over limit, Offline, Bad data, Awaiting data, Not monitored.

COLUMN	UNIT	MEANING AND HOW IT IS DERIVED
Updated	time	Elapsed time since the last reading — the staleness indicator.

Table 7.1 — Point table columns. IMPL

Sorting Any column heading sorts. The default is severity descending, so problems rise to the top without interaction. Re-sorting is suppressed while the pointer or keyboard focus is inside the table, so rows do not move under the reader. IMPL

What to look for High *% of limit*; large positive *vs typical*; any non-Normal status; an *Updated* value that has stopped advancing.

7.3.4 Point detail panel

Purpose Everything known about one location, opened by clicking a row or pressing Enter on it.

Contents A trend chart with warning, alarm and rated levels drawn as labelled reference lines; fifteen figures including smoothed and unsmoothed values, typical, deviation, rated demand, both limits, percentage of limit, session peak and time, session energy, unit count and unit rating, group, feeding substation, service state and sample count; and that point's own alarm episode history. IMPL

Interpretation Seeing the limits drawn on the same axes as the trace answers "how close is this to alarming?" without arithmetic.

Accessibility Opens with focus on the close control; Escape closes.

7.4 How it works

All 110 rows are built once at start-up and thereafter updated in place; filtering toggles row visibility rather than rebuilding the table. Sorting reorders existing nodes via a document fragment. This keeps a 110-row table updating once per second without flicker. IMPL

7.5 Practical use case

Following the Overview alarm, an engineer opens this screen, types "tower" into search, and finds ADM-TWR-C6C7 at 267 kW against a typical of 216 kW — 24% above normal, 109% of its warning limit. Opening the row shows the trend crossing the warning line about eight minutes ago and remaining above it. The alarm history shows no prior episode for this point, indicating a new condition rather than a recurring one.

CHAPTER 8

Screen Analysis — Voltage & Frequency

Supply quality: whether the electricity arriving is good, as distinct from how much of it there is.

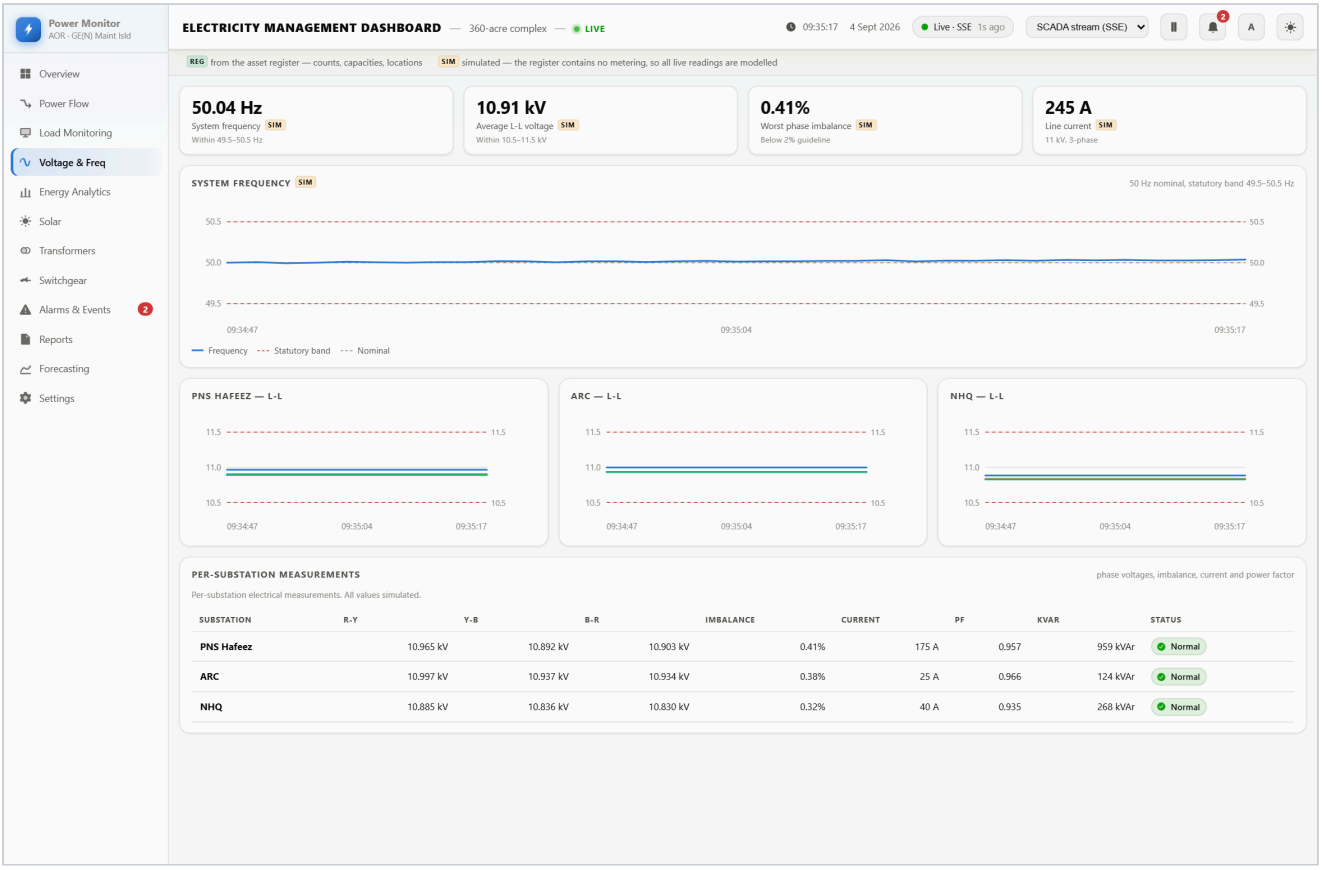


Figure 8.1 — The Voltage & Frequency screen: stat tiles, the frequency trace, three per-substation voltage charts, and the measurements table.

8.1 Purpose

To present the parameters that describe the *quality* of supply — frequency, voltage, phase balance and current — against their statutory limits.

8.2 Why this screen exists

Poor power quality damages equipment and wastes money without ever causing an outage. Voltage outside band shortens insulation life and overheats motors; sustained phase imbalance destroys three-phase motors. These conditions are invisible on any consumption view, so they require their own screen. `DOMAIN`

8.3 Sections

8.3.1 Stat tiles

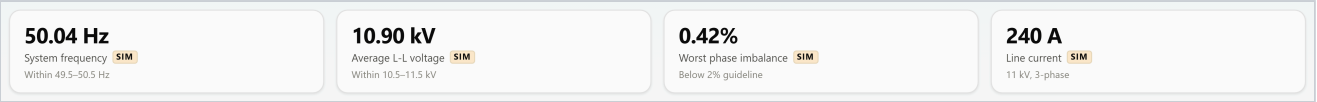


Figure 8.2 — System frequency, average L-L voltage, worst phase imbalance and line current. Each states whether it lies inside its band.

Worst phase imbalance — Given its own headline because it is the parameter that causes the most expensive damage. Calculated as the maximum deviation of any phase from the three-phase mean, expressed as a percentage of that mean, taken across all three substations. The 2% threshold is a standard guideline. `IMPL` `DOMAIN`

8.3.2 System frequency chart

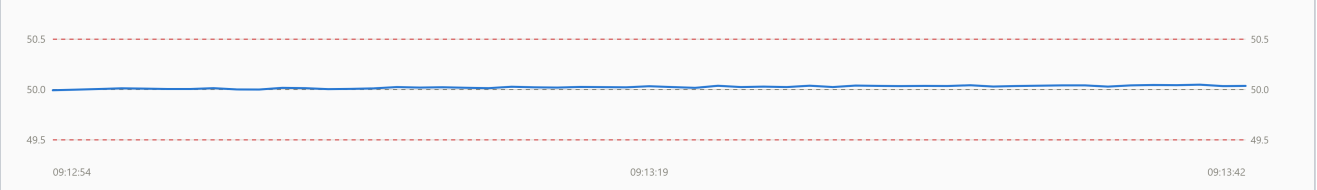


Figure 8.3 — Frequency over time with the 49.5 Hz and 50.5 Hz statutory limits and the 50.0 Hz nominal drawn as labelled reference lines.

Unit — Hz, two decimal places.

Interpretation — Frequency is a national quantity, not a site one. It reflects whether generation matches demand across the grid. A site cannot influence it; it can only respond to it. Sustained excursion beyond the band is a grid event, not a site fault. `DOMAIN`

8.3.3 Per-substation voltage charts

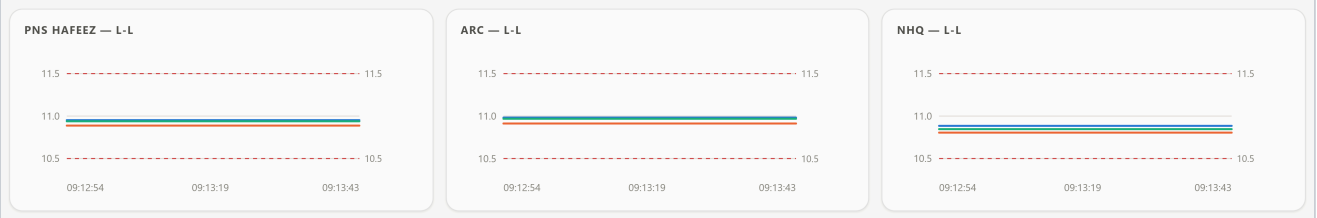


Figure 8.4 — One chart per substation, each showing all three line-to-line phase voltages against the 10.5–11.5 kV band.

Purpose — Localise a voltage problem to a specific substation rather than only knowing the site average.

Interpretation — Three traces moving together indicate a balanced supply. Traces separating indicate imbalance at that substation specifically.

8.3.4 Measurements table

Per-substation electrical measurements. All values simulated.

SUBSTATION	R-Y	Y-B	B-R	IMBALANCE	CURRENT	PF	KVAR	STATUS
PNS Hafeez	10.952 kV	10.888 kV	10.939 kV	0.35%	168 A	0.959	899 kVAr	Normal
ARC	10.982 kV	10.913 kV	10.969 kV	0.38%	26 A	0.966	128 kVAr	Normal
NHQ	10.885 kV	10.805 kV	10.846 kV	0.37%	41 A	0.935	273 kVAr	Normal

Figure 8.5 — Per-substation electrical measurements: all three phase voltages, imbalance, current, power factor and reactive power.

COLUMN	UNIT	MEANING
R-Y, Y-B, B-R	kV	The three line-to-line phase voltages, three decimal places.

COLUMN	UNIT	MEANING
Imbalance	%	Maximum phase deviation from the mean, as a percentage of the mean.
Current	A	Line current derived from power, voltage and power factor.
PF	—	Power factor at that substation, three decimals.
kVAr	kVAr	Reactive power drawn through that substation.
Status	—	Normal when voltage is in band and imbalance is below 2%; otherwise Check.

Table 8.1 — Measurements table columns. `IMPL`

8.4 Practical use case

A maintenance supervisor reports repeated motor failures in one zone. This screen shows the substation feeding that zone running at 2.4% imbalance while the other two sit near 0.9%. That points to unequal single-phase loading on that substation — a wiring distribution problem rather than a motor-quality problem, and one that will keep destroying motors until the loading is rebalanced.

CHAPTER 9

Screen Analysis — Energy Analytics

The shift from power to energy — from what is happening now to what has accumulated. This is the screen for cost conversations.

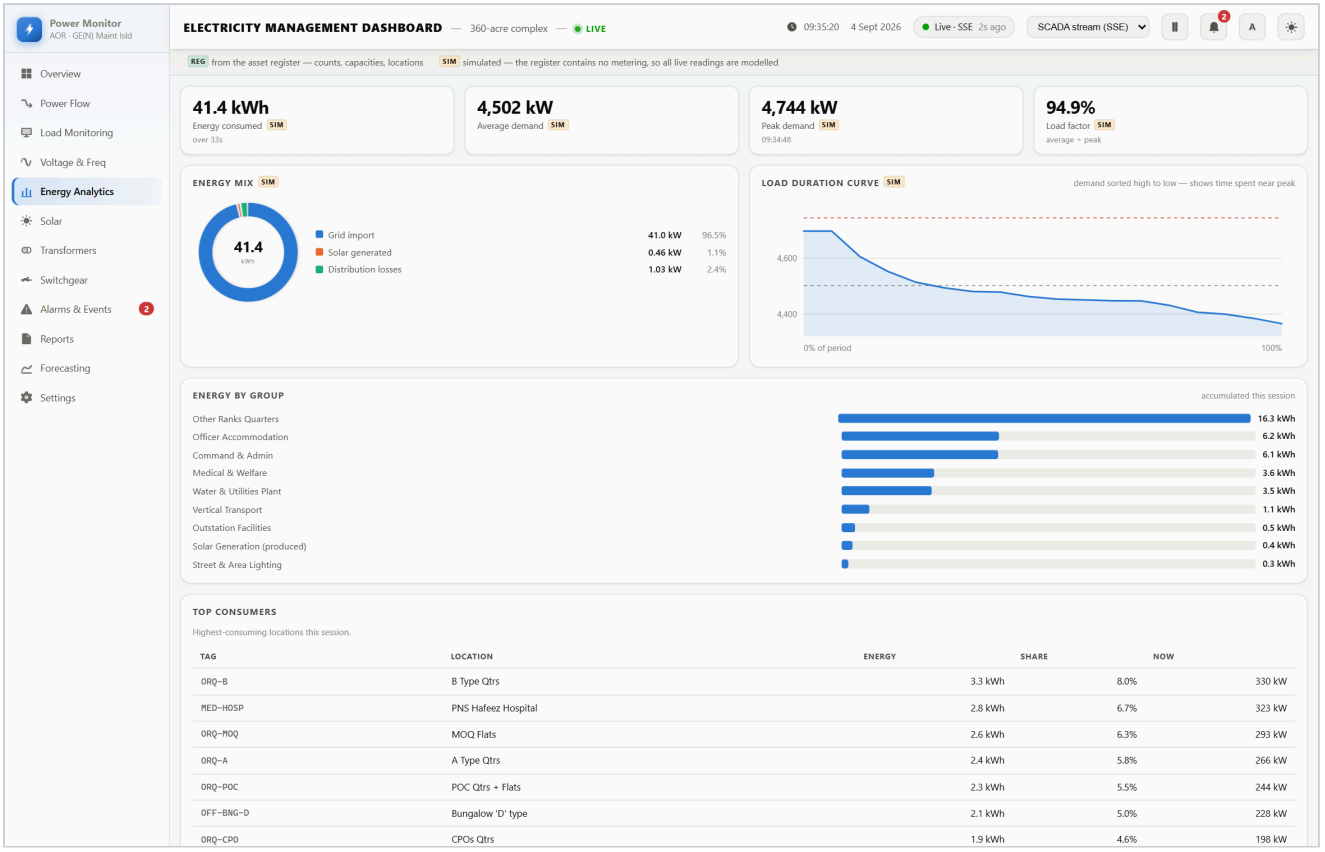


Figure 9.1 — The Energy Analytics screen: stat tiles, energy mix, load-duration curve, energy by group, and top consumers.

9.1 Purpose

To show how much energy has been consumed, how it divides between import, generation and losses, how peaky the demand profile is, and which locations dominate consumption.

9.2 Why this screen exists

Electricity is billed on energy, and frequently on peak demand as well. Every other screen deals in instantaneous power; this one deals in the quantities that appear on an invoice and in a reduction target. `DOMAIN`

9.3 Sections

9.3.1 Stat tiles

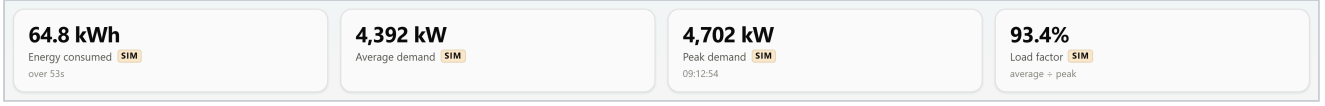


Figure 9.2 — Energy consumed with elapsed period, average demand, peak demand with its timestamp, and load factor.

Average demand	Energy consumed ÷ elapsed hours — the mean power over the period, not a mean of samples. `IMPL`
Load factor	(Average demand ÷ peak demand) × 100. Stated on the tile as "average ÷ peak". `IMPL`

9.3.2 Energy mix

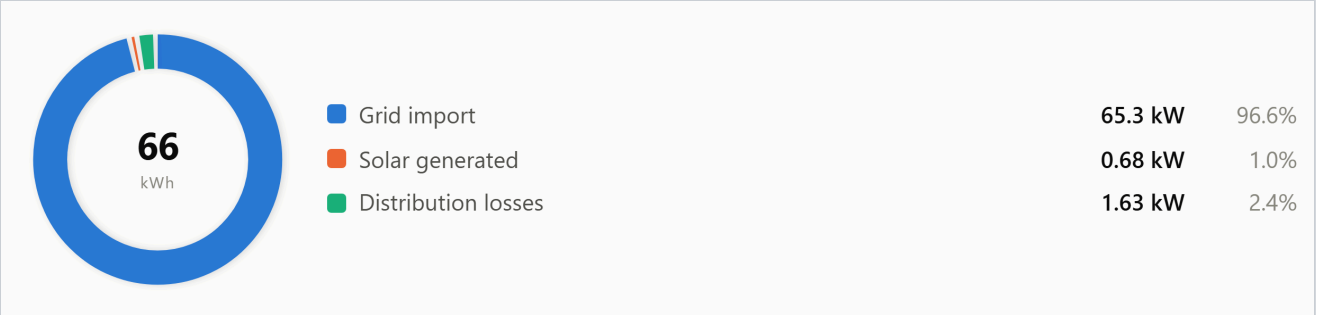


Figure 9.3 — Total energy divided into grid import, solar generation and distribution losses.

Calculation	Grid import = consumed − solar generated; the three segments are import, solar and losses. `IMPL`
Interpretation	The losses segment is energy paid for but never used. Solar directly displaces import.

9.3.3 Load-duration curve

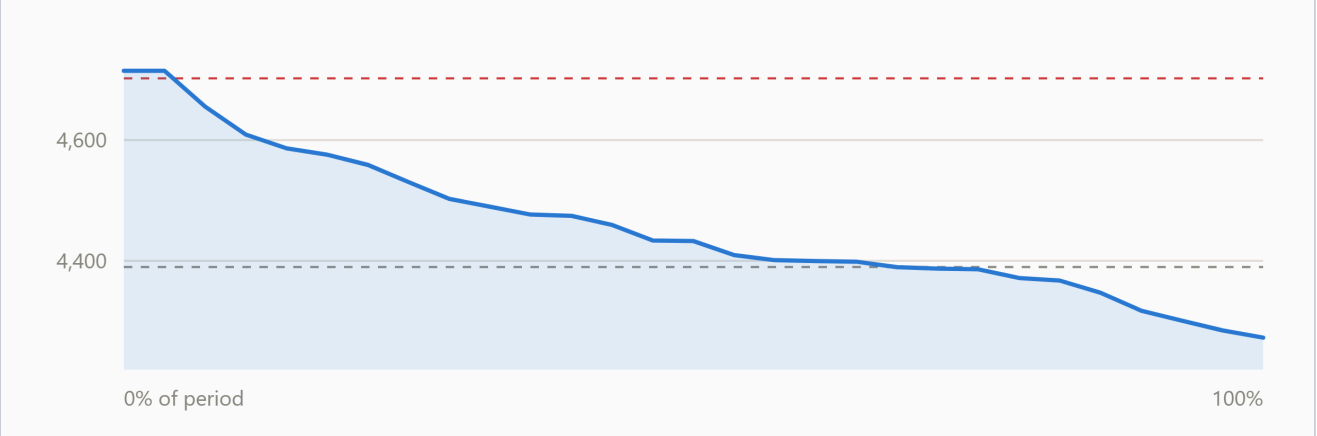


Figure 9.4 — Demand sorted from highest to lowest rather than by time, with peak and average marked as reference lines.

Purpose	Answer a question an ordinary trend cannot: what proportion of the period is spent near peak?
X-axis	Percentage of the period, 0% to 100% — not time of day.
Y-axis	Demand in kW.
Interpretation	A steep drop at the left means peaks are brief and rare, so peak-shaving would yield large savings for little disruption. A flat curve means demand is steady and the peak charge is unavoidable without reducing base load. `DOMAIN`
Why this chart type	It is the standard instrument for peak-demand analysis in energy management, and it is the only chart on the dashboard where the x-axis is not time.

9.3.4 Energy by group

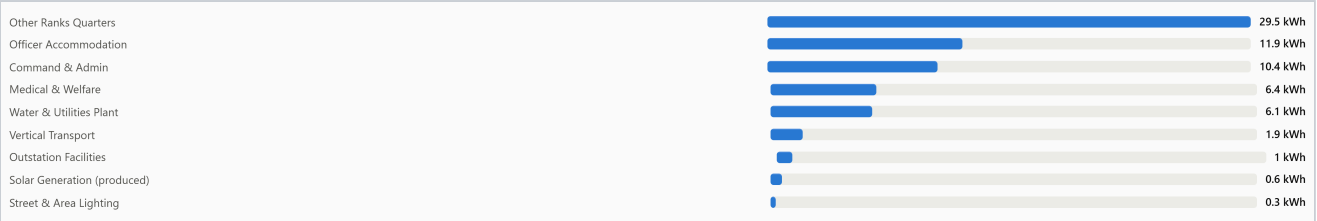


Figure 9.5 — Accumulated energy per asset group, ranked. The generation group is labelled "(produced)" and reported positive.

9.3.5 Top consumers

Highest-consuming locations this session.

TAG	LOCATION	ENERGY	SHARE	NOW
ORQ-B	B Type Qtrs	4.9 kWh	7.2%	291 kW
MED-HOSP	PNS Hafeez Hospital	4.7 kWh	7.0%	309 kW
ORQ-A	A Type Qtrs	4.6 kWh	6.9%	254 kW
ORQ-MOQ	MOQ Flats	4.4 kWh	6.5%	265 kW
OFF-BNG-D	Bungalow 'D' type	4.2 kWh	6.2%	258 kW
ORQ-POC	POC Qtrs + Flats	4.1 kWh	6.1%	247 kW
ORQ-CPO	CPOs Qtrs	3.3 kWh	4.9%	196 kW
ADM-TWR-C6C7	Tower C-6 + C-7	3.1 kWh	4.6%	206 kW
ORQ-POS	POS Qtrs	3 kWh	4.5%	190 kW
ADM-ZAFAR	PNS Zafar Offices / Barracks	3 kWh	4.5%	198 kW

Figure 9.6 — The ten highest-consuming individual locations, with each one's share of site total and its current draw.

Purpose	Identify where consumption-reduction effort produces the largest return.
Calculation	Share = point energy ÷ site energy × 100. Generation points are excluded. `IMPL`
Interpretation	Expect large aggregate tags (MOQ Flats, B Type Qtrs) to dominate, because each represents hundreds of dwellings. A single-unit tag appearing high in this list is the more interesting signal.

9.4 Practical use case

A facilities manager is asked to cut electricity cost by 10%. The load-duration curve shows a steep left-hand drop — demand is near peak for only a small share of the period — so peak-shaving is worth pursuing. Top consumers shows the two largest residential tags accounting for roughly a third of site energy, so tariff or behavioural measures there will move the total more than any plant-side intervention.

CHAPTER 10

Screen Analysis — Solar

The only screen concerned with electricity being produced rather than consumed.

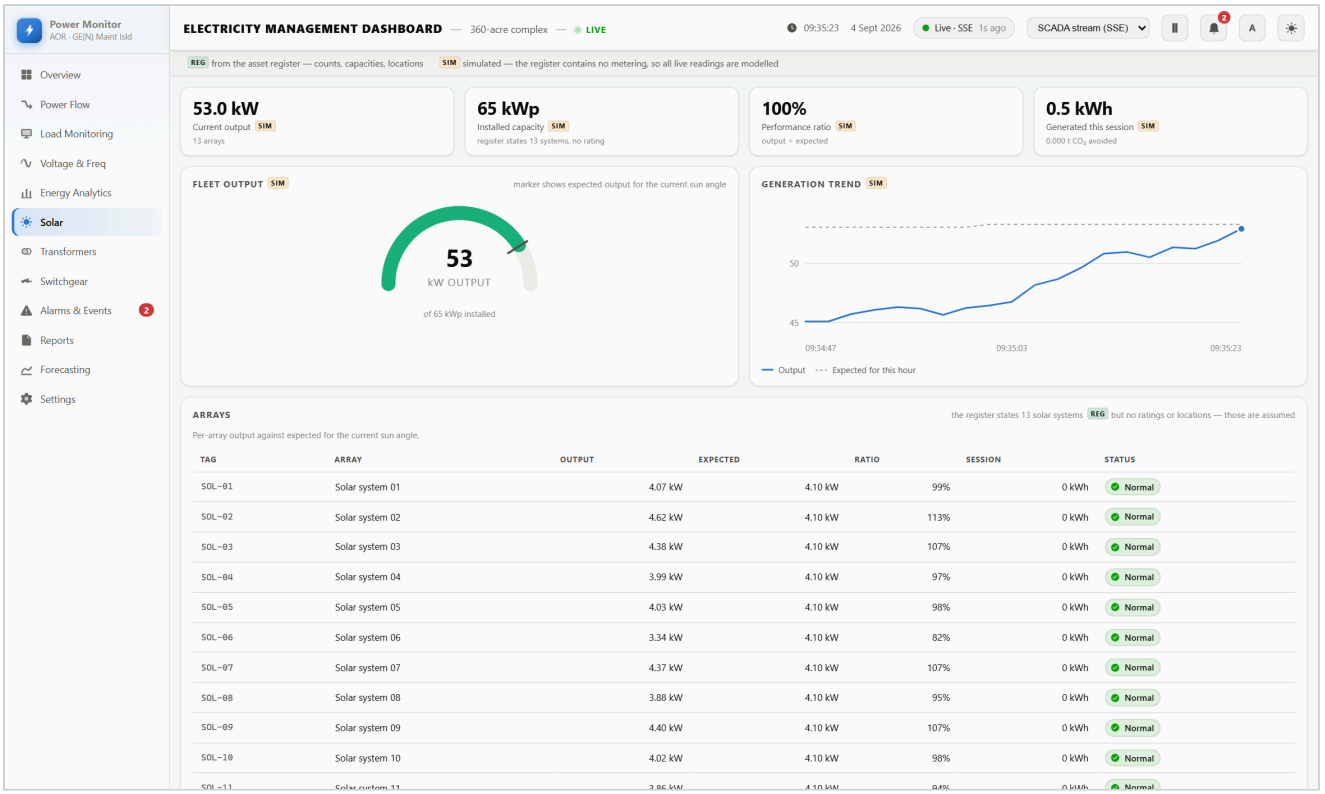


Figure 10.1 — The Solar screen: stat tiles, fleet output gauge, generation trend, and the per-array table.

10.1 Purpose

To report generation from the 13 solar systems and, critically, to judge that output against what the current sun angle should produce rather than against a fixed target.

10.2 Why this screen exists

Solar output varies legitimately by an order of magnitude through the day. A raw output figure therefore cannot distinguish a healthy array at dawn from a failed array at noon. Only a ratio against expected output can, which is why this screen is built around performance ratio rather than kW. `DOMAIN`

10.3 Sections

10.3.1 Stat tiles

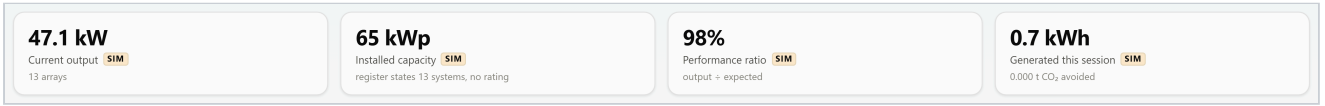


Figure 10.2 — Current output, installed capacity, performance ratio, and session generation with CO_2 avoided. Note the installed-capacity caption records that the register states 13 systems but gives no rating.

Performance ratio	Output ÷ expected × 100, computed only when expected exceeds 0.5 kW; below that the ratio is meaningless and an em-dash is shown. `IMPL`
CO_2 avoided	Session solar energy × 0.42 kg/kWh, reported in tonnes. `IMPL`

10.3.2 Fleet output gauge

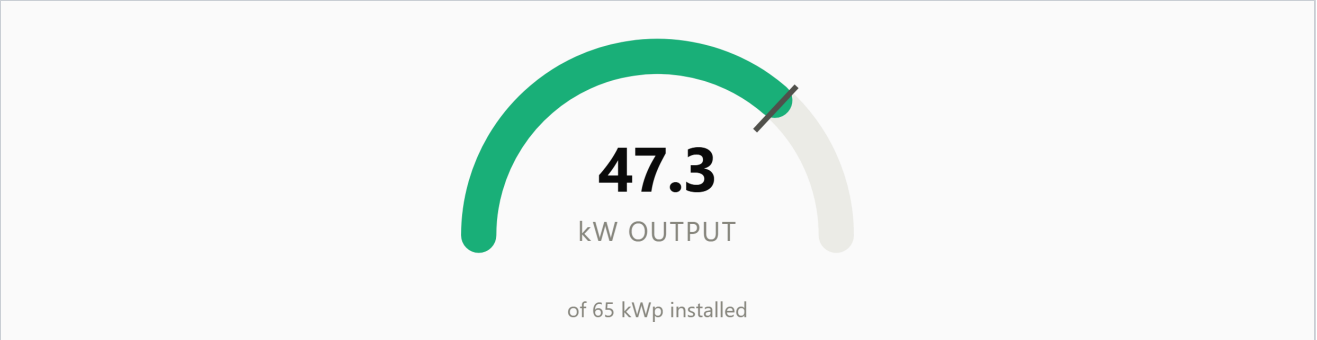


Figure 10.3 — Output against installed capacity. The tick mark on the arc is expected output for the current sun angle; the caption states the installed total.

The gauge should normally read well below full scale — full scale is peak capacity in ideal sun. The meaningful comparison is between the fill and the tick mark, not between the fill and the end of the arc.

10.3.3 Generation trend

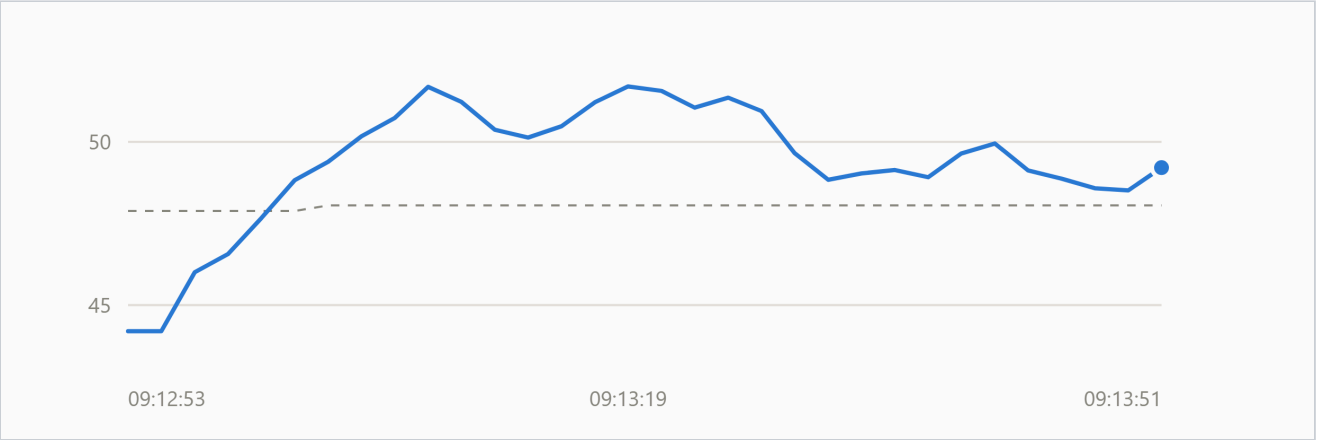
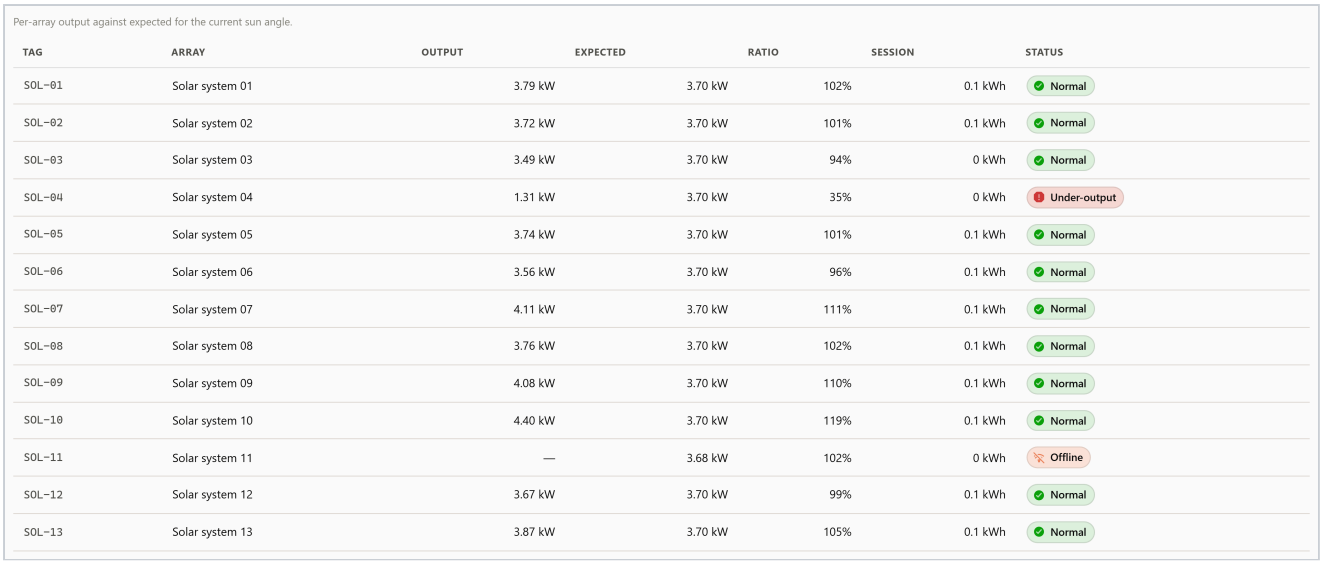


Figure 10.4 — Fleet output over time against expected output. The two traces should track; a persistent gap indicates fleet-wide under-performance.

> **IMPLEMENTATION NOTE**
>
> The trend is derived by scaling one array's history by the number of arrays, rather than by summing all thirteen histories. The shape is representative and the magnitude is correct in aggregate, but it will not reflect divergence between individual arrays. Per-array divergence is visible in the table beneath instead. `IMPL`

10.3.4 Array table



Per-array output against expected for the current sun angle.

TAG	ARRAY	OUTPUT	EXPECTED	RATIO	SESSION	STATUS
SOL-01	Solar system 01	3.79 kW	3.70 kW	102%	0.1 kWh	Normal
SOL-02	Solar system 02	3.72 kW	3.70 kW	101%	0.1 kWh	Normal
SOL-03	Solar system 03	3.49 kW	3.70 kW	94%	0 kWh	Normal
SOL-04	Solar system 04	1.31 kW	3.70 kW	35%	0 kWh	Under-output
SOL-05	Solar system 05	3.74 kW	3.70 kW	101%	0.1 kWh	Normal
SOL-06	Solar system 06	3.56 kW	3.70 kW	96%	0.1 kWh	Normal
SOL-07	Solar system 07	4.11 kW	3.70 kW	111%	0.1 kWh	Normal
SOL-08	Solar system 08	3.76 kW	3.70 kW	102%	0.1 kWh	Normal
SOL-09	Solar system 09	4.08 kW	3.70 kW	110%	0.1 kWh	Normal
SOL-10	Solar system 10	4.40 kW	3.70 kW	119%	0.1 kWh	Normal
SOL-11	Solar system 11	—	3.68 kW	102%	0 kWh	Offline
SOL-12	Solar system 12	3.67 kW	3.70 kW	99%	0.1 kWh	Normal
SOL-13	Solar system 13	3.87 kW	3.70 kW	105%	0.1 kWh	Normal

Figure 10.5 — All thirteen arrays with output, expected, ratio, session energy and status.

COLUMN	UNIT	MEANING
Tag / Array	—	SOL-01 to SOL-13 and the array name.
Output	kW	Current generation; em-dash when offline.
Expected	kW	Rated capacity × the solar profile factor for the current hour.
Ratio	%	Output ÷ expected. The diagnostic column.
Session	kWh	Energy generated since monitoring began.
Status	—	Normal, Low (below 60% of expected), Under-output (below 35%), or Offline.

Table 10.1 — Array table columns. `IMPL`

Under-output limits are evaluated only during daylight, defined as solar profile factor above 0.2. Outside that window the limits are suspended entirely, so no array alarms overnight for the crime of darkness. `IMPL`

10.4 Practical use case

At midday the fleet reads 41 kW against 58 kW expected — a 71% performance ratio. The array table shows twelve arrays near 95% and SOL-07 at 12%, flagged Under-output. The problem is one array, not the fleet: shading, soiling or an inverter fault at that location.

CHAPTER 11

Screen Analysis — Transformers

The only screen containing no simulated values whatsoever.

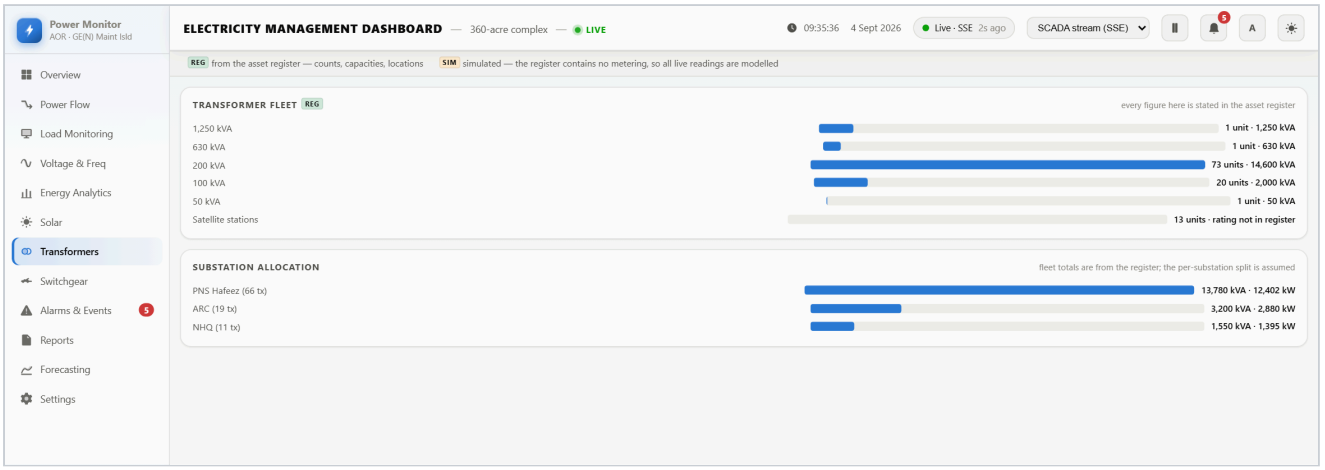


Figure 11.1 — The Transformers screen: fleet composition by unit size, and allocation across the three substations.

11.1 Purpose

To document the transformer fleet: how many units of each rating exist, and how installed capacity distributes across the substations.

11.2 Why this screen exists

Transformer capacity is the ceiling on everything else. Every loading percentage elsewhere on the dashboard is computed against these figures, so they warrant their own screen. It is also the screen to use when every displayed number must be defensible — nothing here is modelled. `IMPL`

11.3 Sections

11.3.1 Transformer fleet

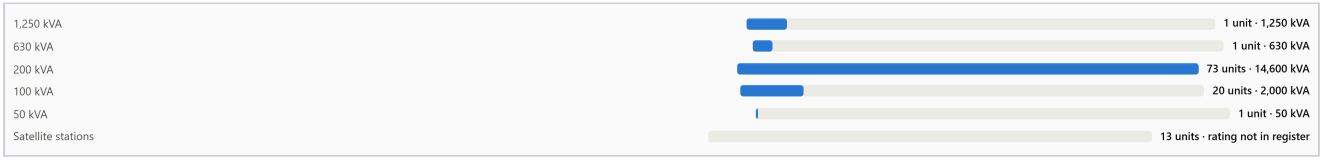


Figure 11.2 — Fleet by unit rating. Satellite stations appear with their count and an explicit note that the register gives no rating for them.

RATING	UNITS	TOTAL KVA	SOURCE
1,250 kVA	1	1,250	Register
630 kVA	1	630	Register
200 kVA	73	14,600	Register
100 kVA	20	2,000	Register

RATING	UNITS	TOTAL KVA	SOURCE
50 kVA	1	50	Register
Satellite stations	13	not stated	Register (count only)
Total	**109**	**18,530**	Reconciles exactly

Table 11.1 — Transformer fleet. Both the unit count and the installed kVA reconcile exactly to the totals stated in the source document; this is asserted by the verification script on every run. IMPL

11.3.2 Substation allocation

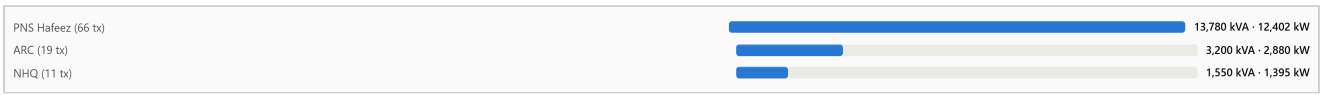


Figure 11.3 — Installed kVA and usable kW per substation. The card subtitle states that fleet totals are from the register while the split between substations is assumed.

SUBSTATION	UNITS	INSTALLED KVA	USABLE KW	RATED DEMAND	UTILISATION
PNS Hafeez	66	13,780	12,402	5,013	40%
ARC	19	3,200	2,880	1,062	37%
NHQ	11	1,550	1,395	741	53%
Satellite	13	not stated	—	100	—

Table 11.2 — Substation allocation. Usable kW = installed kVA × 0.9 assumed power factor. Utilisation figures are typical distribution-transformer loadings. IMPL

> VERIFIED FACT AND STATED ASSUMPTION, SIDE BY SIDE
>
> **Verified:** 109 units and 18,530 kVA are stated in the register and reconcile exactly.
> **Assumed:** which units sit at which substation. The register does not say. The allocation used was chosen so that each substation's utilisation lands in a realistic band, and it still sums exactly to the documented fleet. It must be corrected from the site single-line diagram before being relied upon. IMPL

CHAPTER 12

Screen Analysis — Switchgear

The protective equipment between supply and load.

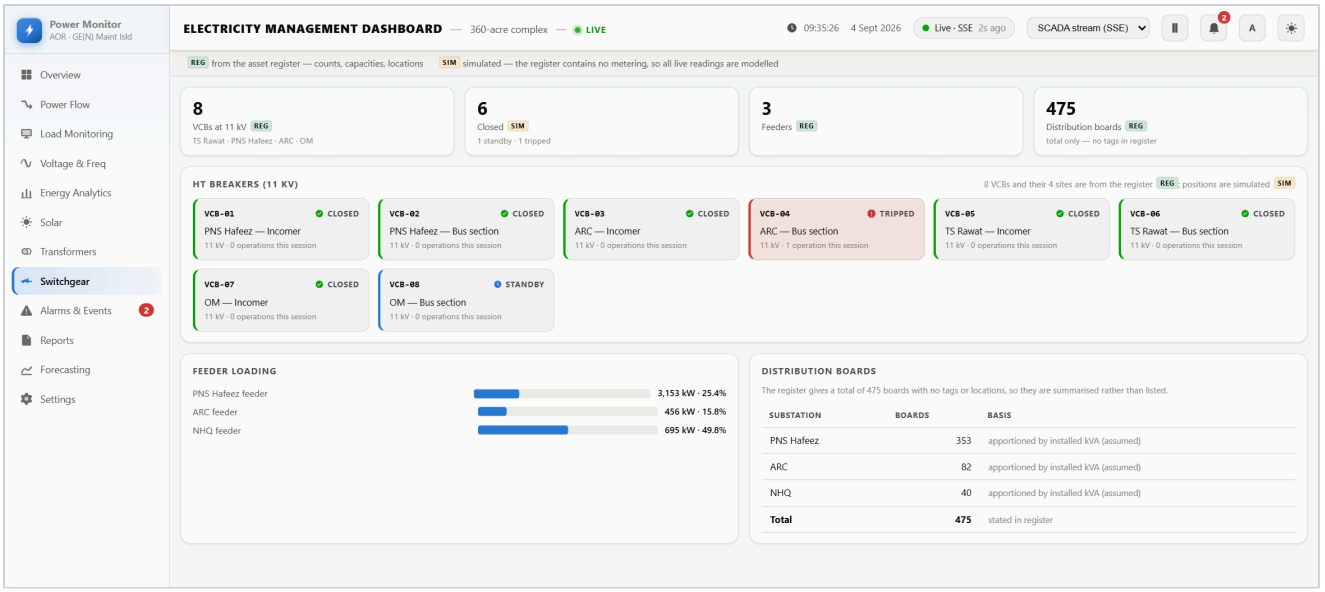


Figure 12.1 — The Switchgear screen: stat tiles, the HT breaker status grid, feeder loading, and the distribution-board summary.

12.1 Purpose

To show the state of the circuit breakers protecting the network, and to summarise the distribution-board population.

12.2 Why this screen exists

Breaker position is the first thing an engineer checks when a section of a site loses supply. A tripped breaker is both the symptom and the location of a fault. `DOMAIN`

12.3 Sections

12.3.1 Stat tiles

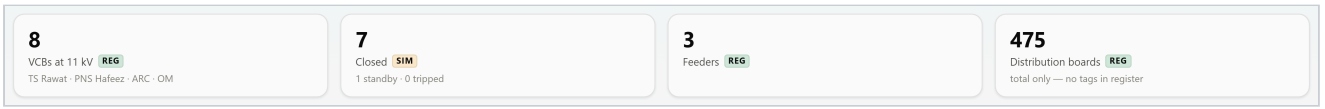


Figure 12.2 — VCB count with the four register-named sites, position summary, feeder count, and distribution-board total. Three of the four carry **REG** badges.

12.3.2 HT breaker grid

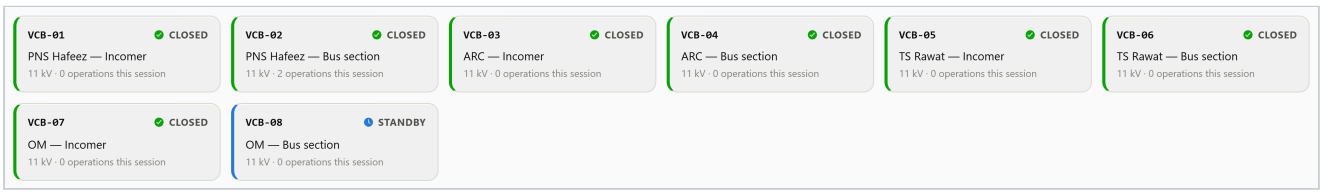


Figure 12.3 — The eight 11 kV breakers, one card each, showing identifier, site and role, position, rating and operation count.

POSITION	MEANING
Closed	Normal operating state — the breaker is carrying power.
Standby	Deliberately open and healthy, available as a spare path.
Open	Open by intent, for example during maintenance.
Tripped	Opened automatically in response to a fault. Requires investigation.

Table 12.1 — Breaker positions. `IMPL`

> PROVENANCE OF THIS SECTION
>
> The register states *"VCBs HT 11 KV — 8 Nos — (TS Rawat, PNS Hafeez, ARC & OM)"*. The count and the four site names are therefore real. Splitting them into incomer/bus-section pairs is an assumption, because the document names the sites without stating what each breaker does. Positions are simulated entirely; the register records no breaker state. `IMPL`

12.3.3 Distribution boards

The register gives a total of 475 boards with no tags or locations, so they are summarised rather than listed.

SUBSTATION	BOARDS	BASIS
PNS Hafeez	353	apportioned by installed kVA (assumed)
ARC	82	apportioned by installed kVA (assumed)
NHQ	40	apportioned by installed kVA (assumed)
Total	**475**	stated in register

Figure 12.4 — The 475 boards summarised by substation, each row stating its basis, with the register total shown for reconciliation.

The boards are deliberately **summarised rather than listed**. The register gives a total of 475 with no names, tags or locations; generating 475 individual board entries would imply a level of detail the source document does not contain. The apportionment shown is by installed kVA and each row says so. `IMPL`

12.4 Practical use case

A zone reports loss of supply. This screen shows VCB-03 (ARC incomer) reading Tripped with one operation this session. The fault is localised immediately to the ARC incomer, and the operation count distinguishes a single event from repeated tripping.

CHAPTER 13

Screen Analysis — Alarms & Events

Everything currently wrong, and a permanent record of everything that has been wrong.

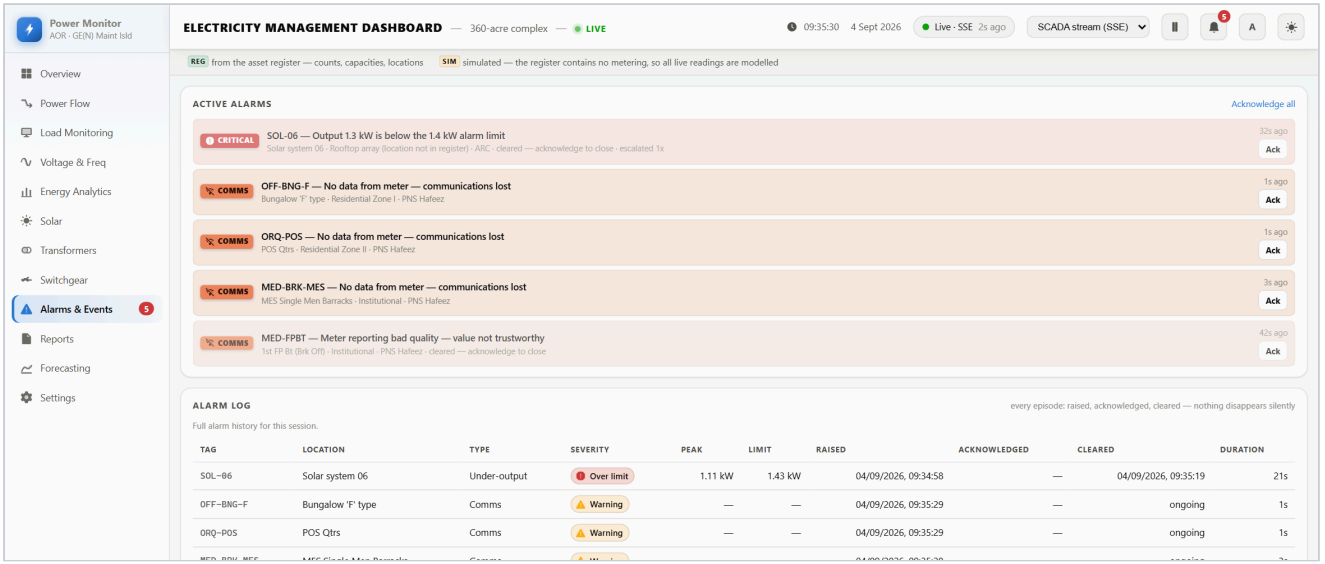


Figure 13.1 — The Alarms & Events screen: the full active list above the complete alarm log.

13.1 Purpose

To hold the complete alarm state and history, with the timestamps needed to answer post-incident questions.

13.2 Why this screen exists

The Overview shows active alarms but is space-constrained and holds no history. After any incident the question asked is not "what is wrong now?" but "when did we know, and how quickly did we respond?" — which requires a durable record. `INTERP`

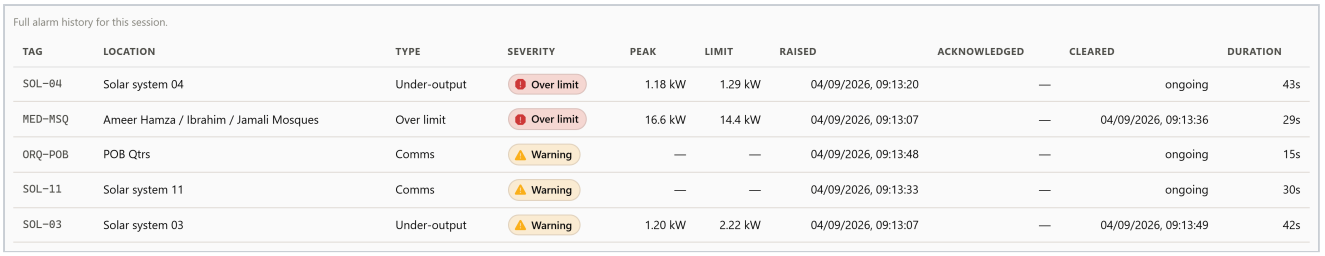
13.3 Sections

13.3.1 Active alarms



Figure 13.2 — The active list, uncapped in height, with an "Acknowledge all" control.

13.3.2 Alarm log


Full alarm history for this session.

TAG	LOCATION	TYPE	SEVERITY	PEAK	LIMIT	RAISED	ACKNOWLEDGED	CLEARED	DURATION
SOL-04	Solar system 04	Under-output	Over limit	1.18 kW	1.29 kW	04/09/2026, 09:13:20	—	ongoing	43s
MED-MSQ	Ameer Hamza / Ibrahim / Jamali Mosques	Over limit	Over limit	16.6 kW	14.4 kW	04/09/2026, 09:13:07	—	04/09/2026, 09:13:36	29s
ORQ-POB	POB Qtrs	Comms	Warning	—	—	04/09/2026, 09:13:48	—	ongoing	15s
SOL-11	Solar system 11	Comms	Warning	—	—	04/09/2026, 09:13:33	—	ongoing	30s
SOL-03	Solar system 03	Under-output	Warning	1.20 kW	2.22 kW	04/09/2026, 09:13:07	—	04/09/2026, 09:13:49	42s

Figure 13.3 — The alarm log. Every episode with its three timestamps and duration.

COLUMN	MEANING
Tag / Location	The affected point and its name.
Type	*Over limit* (consumption), *Under-output* (generation), or *Comms* (communications loss).
Severity	The highest severity the episode reached, not its severity now.
Peak	The most extreme value during the episode — maximum for high alarms, minimum for low.
Limit	The limit that was breached, in kW.
Raised / Acknowledged / Cleared	The three timestamps. Acknowledged is blank if never acknowledged; Cleared reads "ongoing" while active.
Duration	Cleared time minus raised time, or elapsed time if still active.

Table 13.1 — Alarm log columns. `IMPL`

The log retains up to 500 closed episodes in memory. It is **not persisted**: reloading the page clears it. Export to CSV from the Reports screen before reloading if a record is required. `IMPL`

13.4 Practical use case

A monthly review asks how many limit exceedances occurred and how promptly they were handled. The log gives episode count by type and severity, and the gap between Raised and Acknowledged gives the response time for each.

CHAPTER 14

Screen Analysis — Reports, Forecasting & Settings

The final three screens: period reporting, demand projection, and the provenance record.

14.1 Reports

14.1.1 Purpose and rationale

To present period totals and export the underlying data. This screen serves management rather than operations: the audience wants the trend and the numbers, not the moment.

14.1.2 Session summary

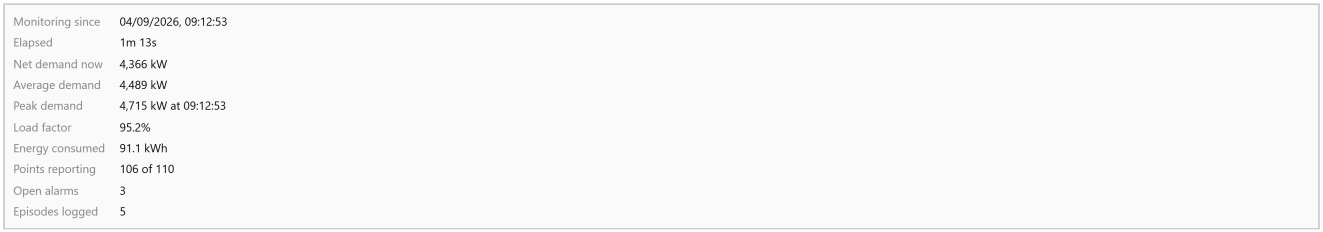


Figure 14.1 — Ten session figures, with CSV export controls beneath.

Monitoring start, elapsed time, demand now, average demand, peak demand with timestamp, load factor, energy consumed, points reporting, open alarms, and episodes logged. `IMPL`

14.1.3 CSV export

EXPORT	COLUMNS
Points CSV	tag_id, name, location, group, zone, substation, units, unit_kw_estimated, rated_kw_estimated, warn_limit_kw, alarm_limit_kw, value_kw, typical_kw, pct_of_limit, severity, quality, session_peak_kw, session_energy_kwh, last_sample_iso
Alarm log CSV	tag_id, name, type, severity, limit_kw, trigger_kw, peak_kw, raised_iso, acked_iso, cleared_iso, duration_s, state, message

Table 14.1 — CSV export schemas. Note that estimated fields are named as such in the column headings, so the distinction survives export into a spreadsheet. `IMPL`

14.1.4 Charts

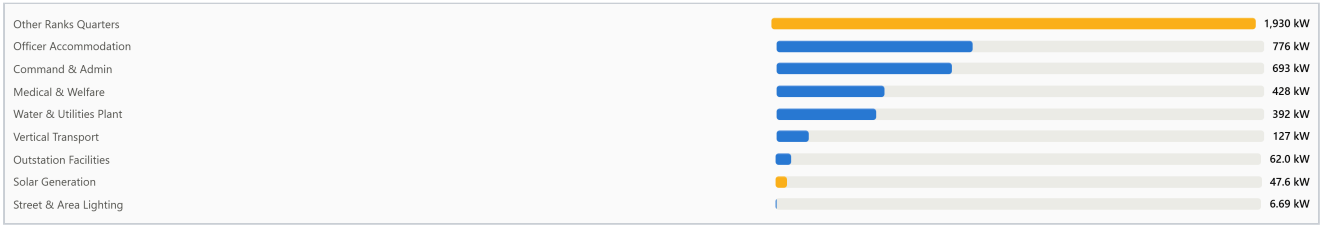


Figure 14.2 — Consumption by group, ranked, with alarm-holding groups coloured.



Figure 14.3 — Substation utilisation against installed capacity, with transformer counts. Amber above 80%, red above 90%.

14.2 Forecasting

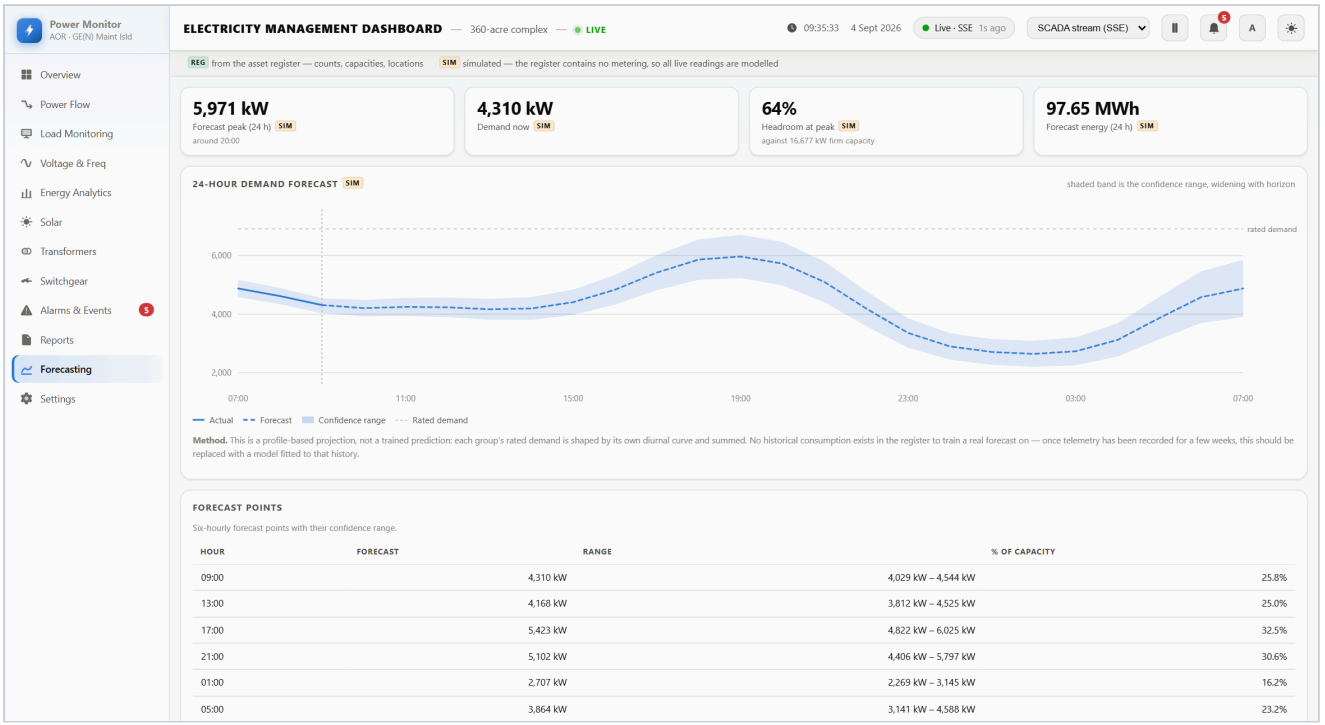


Figure 14.4 — The Forecasting screen: stat tiles, the 24-hour band chart, the method note, and the forecast-points table.

14.2.1 Purpose and rationale

To project demand over the coming 24 hours so peaks can be anticipated and capacity headroom assessed in advance.

14.2.2 Stat tiles

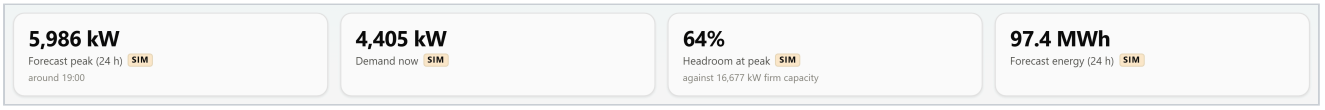


Figure 14.5 — Forecast peak with approximate hour, demand now, headroom at peak, and forecast 24-hour energy.

Headroom = (firm capacity − forecast peak) ÷ firm capacity × 100, where firm capacity is the sum of the three substations' usable kW. `IMPL`

14.2.3 Forecast chart

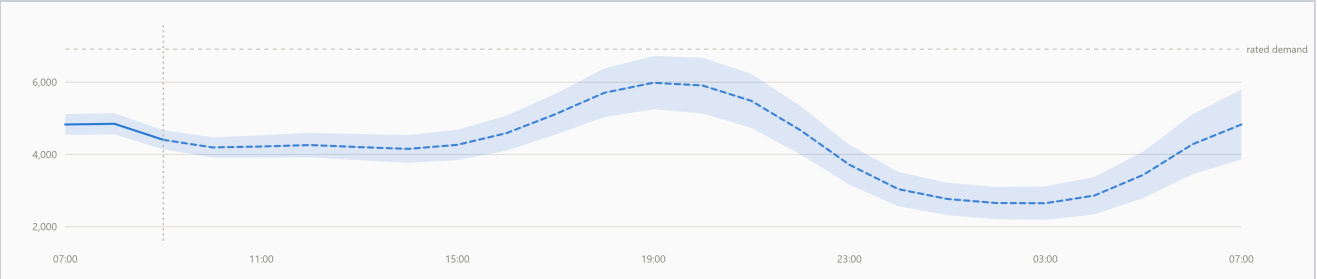


Figure 14.6 — Solid line to the left of the marker is actual; dashed to the right is forecast. The shaded band is the confidence range, widening with horizon.

X-axis	Clock hour, from two hours before now to 22 hours ahead.
Y-axis	Demand in kW, with rated demand marked as a reference.
Band	±6% at the current hour, widening linearly to ±20% at the far horizon. `IMPL`

> METHOD — STATED ON THE SCREEN ITSELF
>
> This is a **profile-based projection, not a trained prediction.** Each asset group's rated demand is shaped by its own diurnal curve and the results summed. No historical consumption exists in the register to train a statistical model on. Once several weeks of real telemetry have been recorded, this should be replaced with a model fitted to that history. The dashboard prints this caveat directly beneath the chart. `IMPL`

14.2.4 Forecast points table

Six-hourly forecast points with their confidence range.

HOUR	FORECAST	RANGE	% OF CAPACITY
09:00	4,405 kW	4,143 kW – 4,672 kW	26.4%
13:00	4,203 kW	3,844 kW – 4,562 kW	25.2%
17:00	5,113 kW	4,546 kW – 5,680 kW	30.7%
21:00	5,484 kW	4,736 kW – 6,232 kW	32.9%
01:00	2,768 kW	2,320 kW – 3,216 kW	16.6%
05:00	3,436 kW	2,793 kW – 4,080 kW	20.6%

Figure 14.7 — Six-hourly forecast values with confidence ranges and each as a percentage of capacity.

14.3 Settings

Point register derived from **AOR - GE (N) Maint Isld.docx**: 2,157 buildings, 475 DBs, 109 transformers (18,530 kVA installed), 38 km HT / 108 km LT cable, 8 VCBs at 11 kV, 3 feeders, 3 substations.

The register contains no metered consumption, voltage, current, power factor, frequency or energy. Every live reading on this dashboard is therefore simulated and badged SIM; unit kW ratings, limits and feeder allocation are engineering estimates. There is no evidence of diesel generators or a 132 kV intake in the register, so neither is modelled. Replace src/register.js and point src/source.js at your SCADA gateway to run on real telemetry.

Figure 14.8 — The full provenance statement: what came from the register, what is estimated, and what is simulated.

This screen carries no controls. It exists to hold the provenance statement and the register gap list in a permanent location — the screen to open if a figure is challenged. Display preferences (theme, density, source) live in the header rather than here. `IMPL`

14.3.1 Practical use case

During a review a stakeholder questions the 16.7 MW connected-capacity figure. Settings shows it derives from 18,530 kVA of installed transformer capacity stated in the register, converted at an assumed 0.9 power factor — and that the per-substation split is an assumption while the total is not.

CHAPTER 15

Filters & Controls

Every interactive control in the interface: what it changes, why it exists, and how it should be used.

15.1 Data filters (Load Monitoring)

CONTROL	TYPE	BEHAVIOUR
Search	Free text	Case-insensitive substring match across tag, asset name and location. Typing "tube" returns all 14 tube wells; "MR-III" returns everything at that location regardless of asset type. Affects the point table only. IMPL
Group filter	Dropdown	Restricts the table to one of the nine asset groups. Kept in sync with the group cards: clicking a card sets this dropdown.
Substation filter	Dropdown	Restricts to points fed by one substation. Useful when a substation-level problem is suspected and the affected loads must be identified.
Only points needing attention	Toggle	Hides everything whose severity is Normal and whose quality is good. Reduces 110 rows to only those in warning, over limit, offline or bad-quality state. Re-evaluated continuously, so a point entering alarm appears without interaction. IMPL
Result count	Readout	Live "*n* of 110 points" so the effect of the active filter combination is always visible.

Table 15.1 — Data filters. All four combine with AND logic. IMPL

> WHAT THESE FILTERS DO NOT DO
>
> There is **no date filter and no time-range selector** anywhere in the dashboard. All views are live plus session history held in memory. Historical range selection would require a historian, which the implementation does not have. This is a genuine functional limitation, not an omission from this report. IMPL

15.2 Table sorting

Every column heading in the point table is a sort control. First click sorts descending for numeric columns and ascending for text; a second click reverses. The active column is marked with an arrow and `aria-sort` for assistive technology. IMPL

Default sort is by severity descending, so the worst condition is always at the top on arrival. Automatic re-sorting is suspended while the pointer or keyboard focus is inside the table, so rows never move while being read.

15.3 Navigation controls

CONTROL	BEHAVIOUR
Sidebar items	Switch the displayed screen. Only the visible screen renders; the choice is remembered in `localStorage` and restored on the next visit. IMPL

CONTROL	BEHAVIOUR
Alarm bell	Jumps directly to Alarms & Events. Carries a badge with the unacknowledged count.
Group cards	Act as filters: select a group and scroll to the table. Clicking the selected card clears the filter without scrolling.
Table rows	Open the point detail panel. Keyboard-reachable; Enter or Space opens, Escape closes.

15.4 Data-source and feed controls

CONTROL	BEHAVIOUR
Source selector	Switches between *Simulator* (in-browser generation), *SCADA stream (SSE)* and *SCADA poll (REST)* at runtime without reload. This is the control that moves the dashboard onto real data. `IMPL`
Pause / resume	Stops the feed so a value can be read or discussed without moving. The connection pill changes to *Paused* so a paused dashboard is never mistaken for a live one.
Acknowledge	Per-alarm button recording that an operator has seen the condition. The timestamp enters the alarm log.
Acknowledge all	Acknowledges every open alarm in one action. Disabled when nothing is unacknowledged.
CSV export	Two buttons on Reports generating point and alarm exports client-side; no server round trip.

15.5 Display controls

CONTROL	BEHAVIOUR
Theme toggle	Switches light and dark palettes, each a separately chosen set of colour steps rather than an inversion. Persisted. `IMPL`
Display density (A)	Increases the root font size from 15 px to 18 px and widens spacing, for a wall-mounted screen read from a distance. Persisted.

CHAPTER 16

KPI & Metrics Reference

The six headline KPIs documented in full, followed by a complete reference table covering every remaining metric in the interface.

16.1 The six headline KPIs

Total Active Power

Definition	The rate at which the site is consuming useful electrical energy at this instant.
Simple	How fast the whole site is using electricity right now — the speedometer, not the odometer.
Technical	Real power: the component of supplied power performing actual work, as opposed to reactive power which circulates without doing work.
Unit	MW on the tile; kW in the sub-line. 1 MW = 1,000 kW.
Purpose	The single most important operational number — the basis of capacity and cost.
Why included	Every other consumption figure is a subdivision of this one.
Data source	Sum of smoothed readings across all monitored non-generation points. `IMPL`
Calculation	`site.kw = Σ smoothed kW of all load points`
Example	4,526 kW displays as 4.53 MW.
Interpretation	Typical 3,000–5,500 kW depending on hour. Judge against *typical for this hour*, not against a fixed threshold — 4,500 kW is normal at 20:00 and abnormal at 04:00.
Relationships	Integrated over time gives Energy. Divided by voltage and power factor gives Current. Combined with reactive power gives apparent power.

Reactive Power

Definition	Power that oscillates between source and load without performing net work.
Simple	Electricity that sloshes back and forth. Motors and transformers need it to build magnetic fields, but it does no useful work and still occupies cable capacity.
Technical	The imaginary component of complex power, arising from the phase displacement between voltage and current in inductive loads.
Unit	MVAR on the tile (megavolt-ampere reactive); kVAr per substation.
Purpose	High reactive power means cables and transformers carry current that produces nothing, and utilities commonly penalise it.
Data source	Derived from real power and power factor per substation, then summed. IMPL
Calculation	`Q = P × tan(arccos(PF))` per substation; site total is the sum.
Example	P = 3,319 kW at PF 0.95 gives Q = 3,319 × tan(arccos 0.95) = 1,091 kVAr.
Interpretation	Rises as power factor falls. Typically 25–35% of real power on a motor-heavy site. A sharp rise with no corresponding rise in real power indicates increased inductive load or failing power-factor correction.
Relationships	Directly determined by power factor; combines with real power to give apparent power.

Power Factor

Definition	The ratio of real power to apparent power — how efficiently supplied power is converted into useful work.
Simple	A score from 0 to 1. 1.0 means every unit supplied does useful work. 0.8 means a fifth of the capacity is wasted moving reactive power around.
Technical	The cosine of the phase angle between voltage and current waveforms.
Unit	Dimensionless, 0 to 1. Displayed to three decimal places.
Purpose	Improving power factor is frequently the cheapest available saving on an industrial site.
Data source	Computed site-wide from real and apparent power. IMPL
Calculation	`PF = P ÷ S` where `S = √(P² + Q²)`
Example	P = 4,526 kW, Q = 1,420 kVAr gives S = 4,744 kVA and PF = 0.954.
Interpretation	**Above 0.95** good. **0.90–0.95** acceptable. **Below 0.90** usually attracts a penalty and the tile shows a warning. Falls as motor load rises — a normal daily variation, not a fault.
Relationships	Governs reactive power and current. For the same real power, a lower power factor means more current and higher losses.

Frequency

Definition	The rate at which alternating current reverses direction.
Simple	How many times per second the electricity changes direction. It must stay very close to 50 — it is a live indicator of whether national generation is keeping up with national demand.
Technical	The fundamental frequency of the supply waveform, determined by the rotational speed of synchronous generators across the interconnected grid.
Unit	Hz (hertz), two decimal places.
Purpose	Sensitive equipment and motor speeds depend on it; sustained excursion signals grid instability.
Data source	Modelled as a slow wander about 50 Hz, identical at every bus because frequency is a system-wide quantity. `IMPL`
Calculation	`f = clamp(50 + 0.045·sin(ϕ) + noise, 49.7, 50.3)`
Interpretation	Statutory band 49.5–50.5 Hz. **Below 49.5** means demand exceeds generation nationally; **above 50.5** the reverse. A site cannot influence this, only respond to it. `DOMAIN`
Relationships	Independent of site load. Unlike every other KPI, it reports on the grid rather than on the site.

Average Voltage (L-L)

Definition	Mean line-to-line voltage across the three substations.
Simple	The electrical "pressure" pushing current through the wires. Too low and motors overheat; too high and insulation degrades.
Technical	RMS potential difference between any two of the three phase conductors, averaged across the phases and then across the substations.
Unit	kV (kilovolts), two decimal places. Nominal 11 kV.
Purpose	Out-of-band voltage shortens equipment life without ever causing an outage.
Data source	Derived per bus with a load-dependent droop and a small independent per-phase offset, then averaged. `IMPL`
Calculation	V_{phase} `= 11 × (1 − loading×0.028 + imbalance + noise)`
Example	Phases at 10.962, 10.905 and 10.875 kV average to 10.91 kV — within band.
Interpretation	Band 10.5–11.5 kV (±4.5%). Voltage falling as load rises is normal droop. Voltage below band at high load indicates an undersized or overloaded supply path.
Relationships	Falls as loading rises; determines current for a given power; underpins the phase-imbalance calculation.

Energy Today

Definition	Total electrical energy consumed since monitoring began.
Simple	The odometer to Total Active Power's speedometer. This is what an electricity bill charges for.
Technical	The time integral of real power over the measurement period.
Unit	kWh below 1,000, MWh above. 1 MWh = 1,000 kWh.
Purpose	The billing quantity and the basis of any consumption-reduction target.
Data source	Accumulated on every ingest from smoothed power and elapsed time. `IMPL`
Calculation	`E += P × Δt ÷ 3,600,000` with Δt in milliseconds, rejecting gaps over 60 s.
Example	4,500 kW sustained for 2 seconds adds 4,500 × 2,000 ÷ 3,600,000 = 2.5 kWh.
Interpretation	Monotonically increasing. Its *rate* of increase is average demand. Judge against the expected-for-elapsed reference on the gauge, not in isolation.
Relationships	Power × time. Divided by elapsed hours gives average demand; with peak demand gives load factor; with the emission factor gives CO_2.

16.2 Complete metrics reference

METRIC	UNIT	CALCULATION	INTERPRETATION
Apparent power	kVA / MVA	$S = \sqrt{(P^2 + Q^2)}$	Total the cables must carry. Why transformers are rated in kVA.
Line current	A	$I = (P \times 1000) \div (\sqrt{3} \times V \times 1000 \times PF)$	Cables are sized by current. Rises if PF or voltage falls at constant power.
Peak / max demand	kW / MW	Running maximum of site demand, with timestamp	Often the basis of a utility demand charge.
Average demand	kW	Energy ÷ elapsed hours	Mean power over the period.
Load factor	%	(Average ÷ peak) × 100	High = steady, efficient. Low = paying for capacity used briefly.
Distribution losses	kWh	Σ P × Δt × lossFraction	Energy lost as heat in cables and transformers. Paid for, never used.
System efficiency	%	(1 − lossFraction) × 100	Share of intake reaching the load. Falls slightly as demand rises.
Loss fraction	—	0.018 + 0.012 × clamp(P÷8000, 0, 1)	1.8% at low load rising to 3.0% at high load.
CO_2 avoided	t	Solar kWh × 0.42 kg/kWh	Emissions displaced by self-generation.
Connected capacity	MW	Σ (installed kVA × 0.9)	Network ceiling. **REG**-derived.
Substation loading	%	(Load ÷ capacity) × 100	20–55% typical. Amber > 80%, red > 90%.
Phase imbalance	%	$\max\lvert V_{ph} - V_{avg}\rvert \div V_{avg} \times 100$	Below 2% acceptable. Sustained excess destroys motors.

METRIC	UNIT	CALCULATION	INTERPRETATION
Typical for this hour	kW	Rated kW × profileFactor(profile, hour)	The reference that makes a raw reading meaningful.
vs typical	%	(Now ÷ typical) − 1	Large sustained positive deviation is the investigation trigger.
% of limit	%	(Now ÷ alarm limit) × 100	Proximity to alarm. The most scannable column in the table.
Rated demand	kW	Unit count × estimated unit kW	**Estimated.** Every limit derives from it.
Warning limit	kW	Rated + (alarm − rated) × 0.55	Early notice, set between rated and alarm.
Alarm limit	kW	Rated × group headroom (1.15–1.40)	The consumption limit proper.
Performance ratio	%	(Solar output ÷ expected) × 100	Distinguishes a dawn array from a failed one.
Headroom at peak	%	(Capacity − forecast peak) ÷ capacity × 100	Whether new load can be connected.
Forecast confidence	kW	v × (1 ± (0.06 + horizon×0.14))	±6% now widening to ±20% at 22 h.
Alarm rate context	count	Open episodes; unacknowledged count	Drives the nav badge and bell.

Table 16.1 — Complete metrics reference. All formulas verified against the implementation. `IMPL`

CHAPTER 17

Technical Terminology

The terms and abbreviations appearing in the interface, each explained first in plain language and then technically. Only terms actually present in the dashboard are included.

17.1 Power and energy terms

kW — kilowatt

Simple: a measure of how fast electricity is being used. A domestic kettle draws about 2 kW. **Technical:** one thousand watts; the SI unit of power, being energy transfer per unit time. **Unit of:** power. **Measured:** in this implementation, reported per point by the feed and summed. **Why it matters:** it is the fundamental quantity every consumption figure is built from. **Interpretation:** a rate, not a quantity — meaningful only at an instant. **Example:** a tube well pump drawing 15 kW.

MW — megawatt

Simple: one million watts, used because site totals in kW become unwieldy. **Technical:** 1 MW = 1,000 kW. **Why it appears:** the KPI tiles display site totals in MW for legibility while retaining kW in the sub-line. **Example:** 4,526 kW shown as 4.53 MW.

kWh — kilowatt-hour

Simple: the amount of electricity used, as opposed to how fast. One kilowatt sustained for one hour. **Technical:** a unit of energy equal to 3.6 megajoules. **Formula:** E (kWh) = P (kW) × t (h). **Why it matters:** this is the billing unit. **Interpretation:** always increasing; its rate of increase is average power. **Example:** a 10 kW pump running 5 hours consumes 50 kWh.

MWh — megawatt-hour

1,000 kWh. The dashboard switches automatically from kWh to MWh above 1,000 kWh to keep the displayed figure readable. IMPL

kVA and MVA — kilovolt-ampere

Simple: the total electrical load a cable or transformer must physically carry, including the part that does no useful work. **Technical:** apparent power, the vector sum of real and reactive power. **Formula:** $S = \sqrt{(P^2 + Q^2)}$. **Why it appears:** transformers are rated in kVA, so all capacity figures in the register are in kVA. **Example:** the site's 109 transformers total 18,530 kVA.

kVAr and MVAR — reactive volt-ampere

Simple: the portion of power that does no work but is still required by motors and transformers. **Technical:** the imaginary component of complex power. **Formula:** Q = P × tan(arccos PF). **Interpretation:** the lower the power factor, the higher this is for the same useful output.

17.2 Supply quality terms

V, kV — volt, kilovolt

Simple: electrical pressure. **Technical:** potential difference. **In this system:** 11 kV high-tension distribution — evidenced by the register's "VCBs HT 11 KV" entry — stepped down to 400 V for buildings. IMPL

L-L — line-to-line

Voltage measured between two phase conductors rather than between one phase and earth. The standard way to quote three-phase voltage, which is why the dashboard labels its voltage figures "L-L".

A — ampere

Simple: how much electricity is flowing. **Technical:** the SI unit of current. **Formula used:** $I = P \div (\sqrt{3} \cdot V \cdot PF)$ for a three-phase system. **Why it matters:** cables are sized by current; exceeding their rating overheats them.

Hz — hertz

Simple: cycles per second. **Technical:** the supply's fundamental frequency, nominally 50 Hz in this region. **Interpretation:** a grid-level quantity that a site observes rather than controls.

PF — power factor

Covered fully in Chapter 16. In brief: the ratio of useful power to total supplied power, between 0 and 1.

R-Y, Y-B, B-R — phase pairs

Simple: industrial supply uses three live conductors, coloured Red, Yellow and Blue. These three labels are the three possible pairings between them. **Technical:** the three line-to-line voltage measurements of a three-phase system. **Why they appear:** the voltage profile plots all three so imbalance between them is visible.

Phase imbalance

Simple: the three phases carrying noticeably different loads. **Formula:** maximum phase deviation from the mean, divided by the mean. **Why it matters:** sustained imbalance overheats and destroys three-phase motors. **Interpretation:** below 2% acceptable.

17.3 Network terms

HT / LT — high tension / low tension

High-voltage distribution (11 kV here) and low-voltage final distribution (400 V). Electricity travels at high voltage because losses fall as voltage rises for the same power. The register records 38 km of HT and 108 km of LT cable. IMPL

Substation

Where high voltage is received and stepped down for local distribution. Three exist: PNS Hafeez, ARC and NHQ, plus 13 satellite stations serving outstations.

Transformer

Simple: the device that changes voltage from high to low. **Technical:** a static machine coupling two windings by a magnetic field. **Rated in:** kVA. **Example:** a 200 kVA unit carries roughly 180 kW of real

load at 0.9 power factor.

Feeder

A cable circuit carrying power from a substation to a group of loads. The register states three feeders; the dashboard models one per substation.

Busbar

The heavy conductor inside a substation to which everything connects — the common junction between incoming supply and outgoing feeders. Drawn as the horizontal line on the single-line diagram.

VCB — vacuum circuit breaker

Simple: an automatic switch that disconnects a circuit when something goes wrong. **Technical:** a breaker interrupting current within a vacuum chamber, which extinguishes the arc cleanly; standard at medium voltage. **In this system:** eight units at 11 kV across four register-named sites. IMPL

DB — distribution board

The panel of protective devices inside a building splitting incoming supply into final circuits — a domestic consumer unit, scaled up. The register records 475 with no further detail.

SLD — single-line diagram

The standard engineering drawing of an electrical network, representing each three-phase circuit as one line. Universally read by electrical engineers, which is why the dashboard uses this form rather than a generic chart.

Solar PV / kWp

Photovoltaic panels generating electricity from sunlight. **kWp** is "kilowatt peak" — output under full reference sun. The register states 13 systems but gives no ratings. IMPL

17.4 Load and capacity terms

Load

Any device or group of devices consuming electricity; also the amount consumed. The dashboard uses it in both senses — "load group" (the devices) and "site load" (the amount).

Demand

Power drawn at a given moment. Essentially interchangeable with load in this context.

Peak demand

The highest demand reached in a period. Commercially significant because utilities frequently levy a charge based on it, independent of total energy. DOMAIN

Connected load vs demand

Connected load is the total that could be switched on at once; **demand** is what is actually drawn. Connected capacity here is 16.7 MW while typical demand is around 4.5 MW.

Diversity

The reason for that gap. Across 2,157 buildings, not everything runs simultaneously, so the real total sits far below the theoretical maximum. This is why networks are not sized to connected load. DOMAIN

Capacity / firm capacity

The load the network can carry safely and continuously, derived here from installed transformer kVA at an assumed 0.9 power factor.

Utilisation

Load as a percentage of capacity. Distribution transformers typically run at 30–60%.

Load profile

The characteristic shape of demand across a day. Residential load peaks in the evening; office load at midday; street lighting runs only at night. Each asset group in the register carries its own profile. IMPL

Generation

Electricity produced on site rather than imported. Here this means solar only — the register records no generators of any kind. IMPL

17.5 Monitoring and alarm terms

SCADA

Full form: Supervisory Control and Data Acquisition. **Simple:** the standard family of industrial systems that collect readings from field equipment and present them centrally. **In this system:** the dashboard is designed to sit on top of one; the reference server emulates its data interface.

Tag / Point ID

The unique code identifying one monitored location, such as `TW-01` or `MED-HOSP`. Everything keys off it: readings, limits, alarms and history.

Telemetry

Measurements transmitted automatically from remote equipment, as opposed to manual reading.

SSE — server-sent events

A one-way streaming protocol over HTTP in which the server pushes updates as they occur. Used in preference to polling because it is lower-latency and lighter on both ends.

REST polling

The alternative: the client requests the current readings on a fixed interval. Simpler for a gateway to implement, at the cost of latency.

EMA — exponential moving average

Simple: a smoothing method weighting recent readings more heavily than older ones. **Formula:** $\alpha = 1 - e^{-\Delta t/\tau}$, then value = previous + α(new − previous), with $\tau = 6$ s. **Why it matters:** raw sensor jitter would otherwise read as plant instability. IMPL

Data quality states

Deliberately a separate axis from severity. **Good** — fresh sample. **Pending** — not heard from yet at start-up, explicitly not an alarm. **Stale** — over 15 s old. **Offline** — over 40 s old; raises a communications alarm. **Bad** — the meter reported bad quality, and limits are not evaluated on it. IMPL

Alarm handling terms

TERM	MEANING IN THIS IMPLEMENTATION
On-delay	A limit must be exceeded continuously for 6 s before an alarm raises, so a single noisy sample cannot trip it.
Hysteresis / deadband	To clear, a value must fall 4% below the limit — not merely back to it. Prevents endless alarm/clear chatter at the threshold.
Off-delay	It must also remain clear for 12 s before the alarm is considered over.
Latching	An alarm persists until acknowledged, even if the condition resolves itself.
Acknowledge	An operator confirming they have seen it; the timestamp is recorded.
Ringback	An alarm that cleared before acknowledgement remains listed, marked cleared, so intermittent faults are not missed.
Escalation	An open alarm worsening updates the same episode rather than raising a second, and cancels any prior acknowledgement.
Re-arm	After an episode closes, the same point cannot re-raise at the same severity for 90 s. The point still *displays* its true condition; only the new annunciation is held.
Alarm flood	The failure mode all of the above guards against — so many alarms at once that operators stop reading them. ISA-18.2 defines it as more than 10 in 10 minutes.
Episode	One complete alarm event from raised through acknowledged to cleared, recorded as a single entry with its peak value.

Table 17.1 — Alarm terminology and the implementation's actual parameters. IMPL

ISA-18.2 and ISA-101

International standards for industrial alarm management and human-machine interface design respectively. The alarm behaviour documented above follows ISA-18.2 practice. DOMAIN

CHAPTER 18

Units & Measurements — Power versus Energy

Every unit appearing in the dashboard, and the distinction that causes more confusion than any other in this domain.

18.1 Units in use

UNIT	NAME	MEASURES	WHERE IT APPEARS
kW	kilowatt	Power	Point readings, limits, group and substation roll-ups, all tables
MW	megawatt	Power	KPI tiles, flow stats, connected capacity, max demand
kWh	kilowatt-hour	Energy	Session energy, per-point energy, energy by group, losses
MWh	megawatt-hour	Energy	Automatic switch above 1,000 kWh
kVA	kilovolt-ampere	Apparent power	Transformer ratings, installed capacity
MVA	megavolt-ampere	Apparent power	KPI sub-line beneath reactive power
kVAr	reactive volt-ampere	Reactive power	Per-substation measurements table
MVAR	mega reactive VA	Reactive power	KPI tile
kV	kilovolt	Voltage	Voltage KPI, voltage profile, per-substation table, SLD
V	volt	Voltage	LT nominal (400 V) in the model constants
A	ampere	Current	Peak current tile, per-substation table, SLD nodes
Hz	hertz	Frequency	Frequency KPI and trend chart
kWp	kilowatt peak	Solar capacity	Installed solar capacity
t	tonne	Mass (CO_2)	CO_2 avoided tile
kg/kWh	—	Emission factor	Constant, 0.42, used for CO_2
%	percent	Ratio	Loading, efficiency, load factor, imbalance, % of limit, vs typical
acres, km, gallons	—	Site data	Register facts: 360 acres, 38 km HT, 108 km LT, tank capacities

Table 18.1 — Complete unit inventory. `IMPL` Note that no currency unit appears anywhere: the dashboard contains no tariff data and performs no cost calculation.

18.2 Conversions used

```
1 MW   = 1,000 kW          1 MWh  = 1,000 kWh
1 MVA  = 1,000 kVA         1 MVAR = 1,000 kVAr
1 kV   = 1,000 V           1 t    = 1,000 kg

Display thresholds implemented:
  energy  switches kWh -> MWh at 1,000 kWh
  power   KPI tiles always MW; tables always kW
  CO2     always reported in tonnes (kg / 1000)
```

18.3 Power versus energy — the central distinction

THE DISTINCTION IN ONE LINE

Power is the *rate* at which electricity is being used, measured at an instant. **Energy** is the *quantity* used over a period. Power is the speedometer; energy is the odometer.

	POWER	ENERGY
What it measures	Rate of use, right now	Total used over a period
Units	W, kW, MW	Wh, kWh, MWh
Car analogy	Speedometer — 90 km/h	Odometer — 450 km travelled
Can it be zero?	Yes, whenever nothing is running	Only before anything has ever run
Direction	Rises and falls continuously	Only ever increases
Billed as	Peak demand charge	Energy charge (the bulk of a bill)
On this dashboard	KPI tile, point table "Now", load trend, flow diagram	Energy Today tile, Energy Analytics, session energy per point

Table 18.2 — Power and energy compared. DOMAIN

18.3.1 The relationship

```
Energy = Power × Time

  Worked example (illustrative):
  A pump draws 10 kW and runs for 5 hours.
  Energy = 10 kW × 5 h = 50 kWh

  As actually implemented, per ingest tick:
  E += P × Δt ÷ 3,600,000        Δt in milliseconds

  Same example over one 2-second tick at 4,500 kW:
  E += 4,500 × 2,000 ÷ 3,600,000 = 2.5 kWh
```

The 3,600,000 divisor converts milliseconds to hours (1 h = 3,600,000 ms). The implementation rejects intervals longer than 60 seconds, so a communications gap does not create a fictitious energy spike. IMPL

18.4 The related quantities

TERM	UNIT	MEANING
Consumption	kWh	Energy drawn by loads over a period.
Generation	kW / kWh	Electricity produced on site. Here, solar only.
Demand	kW	Power drawn at a given moment.
Peak demand	kW	The highest demand reached in a period.
Load	kW	Both the consuming devices and the power they draw.
Capacity	kW / kVA	The maximum the network can carry safely.
Net import	kWh	Consumption minus on-site generation — what the grid actually supplied.

Table 18.3 — Related quantities and their units.

18.5 Precision and formatting rules

The implementation formats numbers to a precision a reader can act on: IMPL

- Values at or above 100 — no decimals (452 kW, not 452.184 kW)
- 10 to 100 — one decimal
- Below 10 — two decimals
- Units always travel with the number, never only in a column heading
- Columns of figures use tabular (fixed-width) digits so they align vertically

CHAPTER 19

Charts, Calculations & Metric Relationships

Every visualisation type, every verified formula, and how the metrics connect to one another.

19.1 Chart inventory

CHART	X-AXIS	Y-AXIS	PURPOSE AND WHAT TO LOOK FOR
Load trend (Overview)	Clock time	kW	Demand shape against typical. Look for a sustained gap above the dashed reference.
Sparkline (table rows)	Sample index	kW	24-sample micro-trend per point, with a reference line at typical. Look for a rising staircase.
Point detail trend	Clock time	kW	One point with warning, alarm and rated lines drawn on. Look for how close the trace runs to the limit lines.
Distribution doughnut	—	—	Share of demand by substation. Structural; expect stability.
Category bars	kW	Group	Ranked group demand. Colour signals alarm state, not magnitude.
Energy gauge	—	—	Energy against expected-for-elapsed. Compare fill to the tick mark, not to full scale.
Single-line diagram	—	—	Network topology with live values. Look for an amber or red substation outline.
Voltage profile	Clock time	kV	Three phases against statutory limits. Look for traces separating.
Frequency trend	Clock time	Hz	Grid frequency against its band. Look for sustained excursion.
Power flow diagram	—	—	Proportional flow, source to group. Compare ribbon thicknesses.
Energy mix doughnut	—	—	Import vs solar vs losses. Look at the losses share.
Load-duration curve	% of period	kW	Time spent near peak. A steep left edge means peaks are brief.
Solar gauge and trend	Clock time	kW	Output against expected. Look for a persistent gap.
Forecast band chart	Clock hour	kW	24 h projection with confidence range. Look at the peak and its headroom.
Status grid (breakers)	—	—	Per-device position cards. Look for anything not Closed or Standby.
Horizontal bars	Value or %	Category	Used for feeders, transformer fleet, energy by group and utilisation.

Table 19.1 — All sixteen visualisation types. `IMPL` Time axes cover the current session only; there is no historical range selection.

19.1.1 Common chart behaviour

- All charts are hand-authored SVG; no charting library is used.

- Line and multi-series charts carry a crosshair and tooltip, reachable by pointer and by keyboard arrow keys.
- Data marks are solid; reference lines (limits, typical) are dashed hairlines with direct labels, because a dashed line here genuinely means "threshold".
- Y-axes zoom to the data range rather than anchoring at zero, which is safe because the marks are lines rather than filled areas. Axes are always labelled so the offset baseline is explicit. IMPL

19.2 Verified calculations

Every formula below was read directly from the implementation. Where a value is a modelling assumption rather than a measurement, that is stated.

19.2.1 Smoothing

```
src/store.js
α = 1 − e^(−Δt / τ)          τ = 6,000 ms
value = previous + α × (new − previous)

Input:  raw kW reading, elapsed ms since the last sample
Output: smoothed kW  |  Why: raw jitter reads as plant instability
Note:   both the smoothed and raw values are retained and displayed
```

19.2.2 Energy accumulation

```
src/store.js (per point and site), src/electrical.js (site)
E += P × Δt ÷ 3,600,000        gaps > 60,000 ms rejected
Input: smoothed kW, elapsed ms   Output: kWh
```

19.2.3 Three-phase electrical derivations

```
src/electrical.js — all derived from real power P
Q  = P × tan(arccos(PF))                  reactive power, kVAr
S  = √(P² + Q²)                          apparent power, kVA
PF = P ÷ S                                 site power factor
I  = (P × 1000) ÷ (√3 × V × 1000 × PF)   line current, A

These are standard three-phase relationships, so the derived
quantities remain mutually consistent by construction.
```

19.2.4 Modelled quantities (assumptions, not measurements)

```
src/electrical.js
loading   = clamp(kW ÷ capacityKw, 0, 1.2)
PF_bus    = clamp(0.975 − loading×0.09 + 0.006·sin(ϕ), 0.82, 0.99)
V_phase   = 11 × (1 − loading×0.028 + imbalance + noise)
frequency = clamp(50 + 0.045·sin(ϕ) + noise, 49.7, 50.3)
lossFrac  = 0.018 + 0.012 × clamp(kW ÷ 8000, 0, 1)     1.8% .. 3.0%
efficiency= (1 − lossFrac) × 100
CO2       = solar kWh × 0.42 kg/kWh
```

Power factor degrading as load rises, and voltage drooping under load, are physically correct behaviours; the specific coefficients are modelling choices, not measured characteristics. INTERP

19.2.5 Register-derived quantities

```
src/register.js
ratedKw       = unit count × estimated unit kW     unit kW is ESTIMATED
alarmKw       = ratedKw × group headroom            1.15 .. 1.40
warnKw        = ratedKw + (alarmKw − ratedKw) × 0.55
capacityKw    = round(installed kVA × 0.9)          0.9 PF assumed

Solar under-output limits are RELATIVE, not absolute:
lowAlarm = expected output × 0.35
lowWarn  = expected output × 0.60
evaluated only when the solar profile factor exceeds 0.2 (daylight)
```

> **WHY THE WARNING LIMIT IS DERIVED FROM THE GAP**
>
> An earlier revision set the warning limit at a flat 85% of the alarm limit. For groups with tight headroom — street lighting at 1.15 — that produced a warning limit *below* rated demand, so those circuits would have alarmed at normal load. Deriving it from the gap between rated and alarm makes that impossible for any headroom value. The verification script asserts the resulting ladder is coherent for every point. IMPL

19.2.6 Analysis calculations

```
src/store.js, src/pages.js
expected      = ratedKw × profileFactor(group profile, hour)
deltaVsExpected = (value ÷ expected) − 1
pctOfLimit    = (value ÷ limit) × 100
avgDemand     = energy kWh ÷ elapsed hours
loadFactor    = (avgDemand ÷ peakDemand) × 100
imbalance     = max|V_phase − V_avg| ÷ V_avg × 100
performanceRatio = (output ÷ expected) × 100      if expected > 0.5 kW
headroom      = (capacity − forecastPeak) ÷ capacity × 100

Forecast, per hour h over a 24-hour horizon:
v(h)     = Σ_profiles ratedByProfile × profileFactor(profile, h)
spread   = 0.06 + max(0, k−2)/22 × 0.14
band     = v × (1 ± spread)
```

19.2.7 Alarm evaluation

```
src/limits.js — two independent stages
STAGE 1  confirm the physical CONDITION (drives the display)
  rise:  severity confirmed after 6,000 ms continuously exceeded
  fall:  clear requires value < limit × (1 − 0.04) for 12,000 ms
         (for low limits: value > limit × 1.04)

STAGE 2  reconcile ANNUNCIATION (drives the alarm list)
  open an episode unless within the 90,000 ms re-arm window
  escalate an existing episode rather than opening a second
  close only when the condition returns to normal

QUALITY  stale > 15,000 ms | offline > 40,000 ms
         start-up grace: never-heard-from is "pending", not an alarm
```

Separating condition from annunciation is what allows a point to *display* as over limit while its alarm is held by the re-arm window — anti-flood machinery must never hide a genuine fault. IMPL

19.3 Metric relationships

19.3.1 Implemented in the application

RELATIONSHIP	FORMULA	WHY IT MATTERS
Power → Energy	$E = P \times t$	Converts the operational quantity into the billing quantity.
Power + PF → Reactive	$Q = P \cdot \tan(\arccos PF)$	Quantifies capacity consumed without doing work.
Power + Reactive → Apparent	$S = \sqrt{(P^2+Q^2)}$	Gives the load transformers must actually carry.
Power + Voltage + PF → Current	$I = P \div (\sqrt{3} \cdot V \cdot PF)$	Current determines cable sizing and heating.

RELATIONSHIP	FORMULA	WHY IT MATTERS
Energy → Average demand	avg = E ÷ hours	Feeds load factor.
Average + Peak → Load factor	LF = avg ÷ peak	Reveals how peaky the profile is.
Load + Capacity → Utilisation	U = load ÷ capacity	Drives loading bars and headroom.
Solar energy → CO_2	CO_2 = kWh × 0.42	Converts generation into an environmental figure.
Rated → Limits	alarm = rated × headroom	Every alarm threshold descends from estimated ratings.
Value + Expected → Deviation	Δ = value÷expected − 1	Turns a raw reading into a judgement.
Output + Expected → PR	PR = out ÷ expected	Separates a dawn array from a failed one.
Load → PF and Voltage	modelled coefficients	Higher loading lowers PF and depresses voltage.

Table 19.2 — Relationships actually implemented. `IMPL`

19.3.2 Domain relationships not implemented

The following are standard in this field but are **absent from this implementation**. They are listed so their absence is explicit rather than assumed:

- **Energy → Cost.** No tariff data exists and no monetary value is calculated anywhere.
- **Consumption → CO_2.** Only *avoided* CO_2 from solar is computed; the emissions of imported energy are not.
- **Peak demand → demand charge.** Peak is tracked but never priced.
- **Degree-days → weather normalisation.** No weather input exists.
- **Harmonics / THD.** Not modelled or displayed.

CHAPTER 20

Data Sources & Data Quality

The origin and status of every class of value the dashboard displays. This is the chapter to consult before citing any figure.

20.1 Classification of every data class

DATA CLASS	STATUS	ORIGIN AND QUALIFICATION
Asset counts	**Real (static)**	Stated in the source register: 2,157 buildings, 980 lights, 84 pumps, 14 tube wells, 475 boards, 13 solar systems, 20 lifts, 16 outstations.
Transformer fleet	**Real (static)**	109 units, 18,530 kVA, with the exact size mix. Reconciles exactly; asserted on every test run.
Named locations	**Real (static)**	Tube wells with locations, four filter plants, three overhead tanks with capacities, sixteen outstations.
Network structure	**Real (static)**	3 substations, 3 feeders, 8 VCBs at 11 kV, 38 km HT, 108 km LT, 360 acres.
Unit kW ratings	**Estimated**	Engineering estimates per asset type. *Not in the register.* Every limit descends from these.
Feeder allocation	**Assumed**	Which loads sit on which feeder is not stated. Must be corrected from the site SLD.
Transformer split	**Assumed**	Per-substation allocation assumed; fleet totals are real.
Circuit splits	**Assumed**	Lighting (980), pumps (84) and vertical transport (20) given only as totals; divided into circuits so they can be metered separately.
All live kW readings	**Simulated**	Generated by `simulator.js` from diurnal profiles with noise, excursions and dropouts.
Voltage, current, PF, Hz, kVAr	**Simulated (derived)**	Computed from simulated power via standard three-phase relationships.
Energy totals	**Calculated from simulated**	Genuine integration, but of simulated power.
Breaker positions	**Simulated**	Device count and four site names are real; positions are not.
Forecast	**Modelled**	Profile-based projection, not a trained prediction.
Alarm episodes	**Real logic, simulated input**	The alarm engine is production-grade; the readings it evaluates are simulated.
Historical data	**None**	Session memory only. No historian, no database, no persistence across reload.

Table 20.1 — Data classification. `IMPL`

> **STATEMENT REQUIRED BY THE ACCURACY RULE**
>
> No part of this dashboard currently displays real-time operational measurement. Every live value is simulated. The dashboard is a fully functional monitoring application connected to a simulated feed, not a monitoring application connected to plant. The distinction is marked in the interface at the point of use and must be preserved in any onward presentation. IMPL

20.2 Simulation parameters

PARAMETER	VALUE	EFFECT
Noise amplitude	0.05	AR(1) innovation as a fraction of rated demand.
Mean reversion	0.18	Pull back toward the profile each tick.
Excursion chance	0.0015	Per point per tick; leaves roughly 5% of points excursing at any moment.
Dropout chance	0.0005	Roughly 3% of points offline at any moment.
Bad-quality chance	0.0003	Under 1% of points reporting bad quality.

Table 20.2 — Simulator event rates. These are per-point per-tick probabilities; the resulting steady-state fraction is approximately probability × (event duration ÷ tick interval). IMPL

20.3 Data-quality handling

The most important design decision in this area is that **quality is a separate axis from severity**. A meter that has stopped reporting is not the same fact as a meter correctly reporting zero, and the two are never conflated. The payload contract requires unreachable meters to be omitted rather than zeroed, and the dashboard surfaces them as a distinct communications condition. IMPL

Feed health is watchdogged independently of message content: *live* → *degraded* after 8 s of silence → *down* after 25 s. When the feed dies a banner states so and the view dims, rather than the display freezing on stale values while appearing healthy.

CHAPTER 21

Reading the Dashboard & Design Rationale

The recommended workflow, and why each class of feature exists.

21.1 Recommended workflow

STEP	ACTION	WHAT YOU ARE ESTABLISHING
1	Overview — KPI strip	Is supply quality acceptable? Four indicators: power factor, frequency, voltage, plus total demand against typical.
2	Overview — alarms	Is any location over limit? Empty means no.
3	Overview — trend and SLD	Is demand tracking its normal shape, and are all substations neutral?
4	Check the connection pill	Confirm the data is current. A paused or lost feed invalidates steps 1–3.
5	If an alarm exists → Load Monitoring	Filter to "only points needing attention"; identify the specific location.
6	Open the point detail	See the trend against its limits and the point's alarm history.
7	If supply-related → Voltage & Frequency	Distinguish a load problem from a supply problem.
8	If structural → Power Flow / Switchgear	Localise to a feeder, substation or breaker.
9	For review → Energy Analytics / Reports	Period totals, consumption drivers, CSV export.
10	For planning → Forecasting	Anticipated peak and capacity headroom.

Table 21.1 — Recommended reading order. INTERP

> THE FOUR-CHECK ROUTINE
>
> For continuous monitoring, steps 1–4 take a few seconds and are sufficient. If the KPI indicators are green, the alarm panel is empty, the trend tracks its reference and the feed is live, no further inspection is warranted.

21.2 Why each feature exists

FEATURE	RATIONALE
KPI tiles	Six parameters characterise the electrical state. Presenting them as large single figures makes the normal state readable from across a room, which is what makes monitoring continuous rather than periodic.

FEATURE	RATIONALE
"Typical for this hour"	A raw figure cannot be judged. 4,500 kW is normal at 20:00 and alarming at 04:00. Every reading carries its reference so a judgement is possible without knowing the site by heart.
Status as icon + word + colour	Status must survive colour-blindness, greyscale printing and glance-reading under stress. Colour alone fails all three.
Three severity tiers	ISA-18.2 recommends no more than three or four. Beyond that, operators stop trusting the distinction.
On-delay and hysteresis	Without them a value sitting on a threshold alarms and clears endlessly, and operators learn to ignore the alarm list.
Latching and acknowledgement	An alarm that disappears unseen erodes trust in the system permanently. Persistence plus a recorded acknowledgement makes response auditable.
Ringback	Intermittent faults are the ones most worth catching and the easiest to miss.
Offline distinct from zero	A dead circuit and a failed communications link demand different responses. Conflating them is a classic and expensive fault in this domain.
Feed watchdog and banner	The most dangerous failure of a live dashboard is silently freezing on stale values while appearing healthy.
Smoothing	Raw sensor jitter reads as instability. Filtering before display and before limit evaluation prevents both false alarm and false alarm about stability.
Single-line diagram	It mirrors how an electrical engineer already thinks about the plant, so no translation from a generic chart is required.
Load-duration curve	Answers "how much of the time are we near peak?", which a time-series trend structurally cannot.
Performance ratio for solar	Absolute output cannot distinguish a healthy array at dawn from a failed one at noon.
Sortable, filterable table	110 locations cannot be scanned. Default severity sort puts problems on top without interaction.
CSV export	Decision-makers frequently need the period trend in a spreadsheet more than another live chart.
REG / SIM provenance badges	Without them, modelled values would carry the same visual authority as measured ones.
Register data-gaps panel	Surfacing documentary discrepancies where they are used prevents an assumption being mistaken for a measurement.
Wall-display density	These dashboards are frequently mounted and read at distance.
Forecasting method note	A forecast that does not disclose its method invites misplaced confidence.

Table 21.2 — Feature rationale. `INTERP`

CHAPTER 22

Technical Findings, Glossary & Conclusion

Observations arising from the analysis, the consolidated glossary, and closing assessment.

22.1 Technical findings

F-1 Gross versus net demand inconsistency

The *Total Active Power* KPI displays gross site load, while the load trend chart directly beneath it plots *net* demand (load minus solar generation) and is labelled "Net demand". At present solar contributes under 1% of site demand, so the two differ by a negligible margin, but the labels describe different quantities. If generation capacity grows the divergence becomes material. IMPL **Recommendation:** align the two, or label the KPI explicitly as gross.

F-2 Solar trend is extrapolated from a single array

The generation trend on the Solar screen scales one array's history by the array count rather than summing all thirteen. Aggregate magnitude is correct and the shape is representative, but divergence between arrays will not appear in that chart. Per-array divergence is visible in the table beneath, so the information is available — but the chart alone would not reveal a single failing array. IMPL

F-3 No persistence

All readings, energy totals and alarm history exist only in browser memory. Reloading the page discards them. For a production deployment a historian would be required, both for record-keeping and to enable the trained forecast that the Forecasting screen currently cannot provide. IMPL

F-4 No authentication or authorisation

Any user reaching the URL has full access, including alarm acknowledgement. Acknowledgement is recorded against the fixed label "operator" rather than an identified user, so the audit trail records that an alarm was acknowledged but not by whom. IMPL

F-5 Every limit rests on estimated ratings

The most consequential finding. No unit kW rating is measured; all are engineering estimates, and every warning and alarm limit derives from them. The alarm logic is sound, but the thresholds it enforces are provisional until real nameplate or metered data replaces the estimates. IMPL

F-6 Documentary discrepancy in the register

The register states 14 tube wells with 12 operational, but lists 11 with locations. One operational tube well is unaccounted for. Raised with the facilities team rather than reconciled. IMPL

F-7 Items that could not be verified

- **Actual per-substation transformer allocation** — not stated in the register; could not be verified from the available implementation or source document.
- **Actual feeder-to-load allocation** — not stated; could not be verified.
- **Satellite station ratings** — the register gives a count of 13 with no kVA; could not be verified.

- **Real breaker roles and positions** — four sites are named but roles are not; could not be verified.
- **Distribution board identities** — 475 stated as a total with no tags or locations; could not be verified.
- **Any historical consumption** — none exists in the source document; could not be verified.

22.2 Consolidated glossary

TERM	FULL FORM	UNIT	SIMPLE MEANING	USE IN DASHBOARD
A	Ampere	A	How much electricity is flowing	Per-substation table, SLD, peak current tile
Alarm limit	—	kW	The threshold that raises an alarm	Point table, detail panel
Apparent power	—	kVA	Total load cables must carry	KPI sub-line, transformer ratings
Capacity	—	kW/kVA	Maximum the network can carry	Loading bars, headroom
DB	Distribution board	—	Panel splitting supply into circuits	Switchgear summary
Demand	—	kW	Power drawn at this moment	Throughout
EMA	Exponential moving average	—	Smoothing that favours recent readings	Applied to all readings
Energy	—	kWh/MWh	Quantity used over time — the odometer	Energy tile, Energy Analytics
Feeder	—	—	Cable circuit from substation to loads	Power Flow, Switchgear
Frequency	—	Hz	Cycles per second; must stay near 50	KPI, frequency chart
Generation	—	kW/kWh	Electricity produced on site	Solar screen, energy mix
HT / LT	High / low tension	kV / V	High- and low-voltage distribution	SLD, register facts
Hysteresis	—	%	Margin required to clear an alarm	Alarm engine, 4%
kVAr	Reactive volt-ampere	kVAr	Power that does no useful work	KPI, per-substation table
kW	Kilowatt	kW	Rate of use — the speedometer	Throughout
kWh	Kilowatt-hour	kWh	Amount used; the billing unit	Energy figures
kWp	Kilowatt peak	kWp	Solar output in full sun	Solar capacity
L-L	Line-to-line	kV	Voltage between two phases	Voltage KPI and profile
Load factor	—	%	Average divided by peak demand	Energy Analytics, Reports

TERM	FULL FORM	UNIT	SIMPLE MEANING	USE IN DASHBOARD
MW / MWh	Mega- watt / watt-hour	MW/MWh	1,000 kW / 1,000 kWh	KPI tiles, large totals
Peak demand	—	kW	Highest demand in the period	Metric tile, Energy Analytics
PF	Power factor	—	Efficiency score from 0 to 1	KPI, per-substation table, SLD
Phase imbalance	—	%	Unequal loading across three phases	Voltage & Frequency
Point / Tag	—	—	Unique ID for one location	Every table and alarm
Performance ratio	—	%	Solar output against expected	Solar screen
R-Y, Y-B, B-R	Phase pairs	kV	The three phase-to-phase voltages	Voltage profile
Rated demand	—	kW	Estimated normal draw for a point	Basis of all limits
SCADA	Supervisory Control and Data Acquisition	—	Industrial monitoring system	The intended data source
SLD	Single-line diagram	—	Standard network drawing	Overview
SSE	Server-sent events	—	Server pushes updates as they happen	Primary transport
Substation	—	—	Where voltage is stepped down	SLD, distribution, Power Flow
Transformer	—	kVA	Changes voltage high to low	Transformers screen
Typical	—	kW	Normal draw for this hour	Every reading's reference
VCB	Vacuum circuit breaker	kV	Automatic protective switch	Switchgear, SLD
Voltage	—	V / kV	Electrical pressure	KPI, voltage profile
Warning limit	—	kW	Early-notice threshold	Detail panel, alarm engine

Table 22.1 — Consolidated glossary, alphabetical.

22.3 Conclusion

Power Monitor is a complete and technically sound monitoring application. Its twelve screens cover situational awareness, per-location detail, supply quality, network structure, energy analysis, generation performance, asset inventory, exception handling, reporting and forecasting. The alarm engine follows recognised industrial practice, with on-delay, hysteresis, latching, acknowledgement, ringback and anti-flood re-arm implemented and covered by automated tests.

Its architecture is deliberately minimal — no framework, no charting library, no database, no runtime dependencies — which makes it deployable by copying a folder onto an isolated network, and gives it

a single well-defined integration seam at `src/source.js`.

Its principal limitation is not one of engineering but of input data. The source register is an asset inventory containing no measurements, so every live value is simulated and every alarm threshold rests on an estimated rating. The implementation handles this honestly: provenance is marked at the point of use, documentary discrepancies are displayed rather than hidden, and the forecast discloses its own method. That candour is the report's basis for judging the system ready for connection to real telemetry.

To move from prototype to operational service, three things are required: real metering data in place of the simulated feed; real nameplate ratings in place of the estimated ones, so the limits become defensible; and a historian, without which no record survives a page reload and no trained forecast is possible.

> **SUMMARY JUDGEMENT**
>
> The application is production-quality in construction and prototype-quality in data. The gap between the two is well documented within the system itself, and closing it is a data-acquisition task rather than a software one.